

Public Available Specification

# **CiA 404**

Version 1.2.0

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## **CANopen profile for measuring devices and closed-loop controllers**

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May 15, 2002



## HISTORY

| Date       | Changes                                                                                                                                                                                                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2002-05-15 | <i>Publication of version 1.2.0 as draft standard (now publicly available)</i><br>NOTE: This document has been converted into "docx format".<br>The conversion caused minor layout differences to the predecessor document in<br>"doc format". The technical content word-by-word is the very same. |

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## **1 SCOPE**

This document represents the device profile for measuring devices and for universal controller devices for measuring or controlling physical quantities like temperature or pressure. These devices use communication techniques which are conform to those described in /2/. This document should be consulted in parallel to this profile.



## 2 REFERENCES

- /1/ ISO 11898, Road vehicles - Interchange of digital information - Controller area network (CAN) for high-speed communication, 1993
- /2/ CiA DS 301 V4.01, CANopen Application Layer and Communication Profile, May 2000
- /3/ CiA DS 401 V2.0, CANopen Device Profile for Generic IO Modules, December 1999
- /4/ CiA DSP 302 V1.0, Framework for Programmable CANopen Devices, April 1998
- /5/ CiA DRP 303-2 V1.0, Representation of SI Units and Prefixes, July 1999

### 3 DEFINITIONS AND ABBREVIATIONS

**CAN**

Controller Area Network. Data link layer protocol for serial communication as specified in /1/.

**CiA**

CAN in Automation international users and manufacturers group e.V. Non-profit association promoting Controller Area Network (CAN).

**COB**

Communication Object, which is made of one or more CAN frames. Any information transmitted via CANopen has to be mapped into COB's.

**COB-ID**

COB-Identifier. Identifies a COB uniquely in a CAN network. The identifier determines the priority of that COB in the data link layer, too.

**I/O**

Input and output

**RPDO**

Receive Process Data Object. Communication object of a device, which contains output data.

**SDO**

Service Data Object. Peer-to-peer communication with access to the Object Dictionary of a CANopen device.

**TPDO**

Transmit Process Data Object. Communication object of a device, which contains input data

## 4 OPERATING PRINCIPLE

### 4.1 Introduction

The purpose of this device profile is to describe units for measuring or closed loop control of different physical quantities. As such devices may include quite different functionality, the profile specifies different functional blocks, which could be combined in a real device. Which functional blocks are included is coded within the device type object (1000h)

The purpose of the devices described in this document is to measure or control a physical quantity in a process environment. Often these types of devices are implemented as multiple channel devices.

Once configured via SDO, the device is able to run its measuring or control tasks without necessarily needing further real time communication with host systems. Every device has a large amount of data for configuration purposes. The control loop itself is realised in the controller device. All data of the device has to be accessed via SDO.

In order to reduce busload the device communicates based on events. In general there are three types of events - information, warnings and errors.

- Information - this event is used for indicating that a set-point has been reached, or that an output has been set. Information events may also be used to transfer data (e.g. measuring values) on a cyclical basis. For this purpose it is suggested to use the SYNC-message specification of [2].
- Warning - this event indicates that a warning limit has been reached,
- Error - this event indicates that an error has occurred on the device.

The event-based communication can be established in the following manner.

1. The device indicates the host device in the network that an event has occurred.
2. The host device reads the data from the device via SDO.

## 4.2 General object dictionary layout

All parameters of the device are accessible via the object dictionary. In order to implement multiple channel devices, parameters are arranged in arrays. Each array has up to `number of channels + 5` entries. The maximum number of available channels is 199.

| Index | Sub-Index | Parameter                                             |
|-------|-----------|-------------------------------------------------------|
| xxxxh | 0h        | number of entries                                     |
|       | 1h        | parameter x for channel 1                             |
|       | :         | :                                                     |
|       | C7h       | parameter x for channel 199                           |
|       | C8h       | reserved                                              |
|       | :         | :                                                     |
|       | FEh       | reserved                                              |
|       | FFh       | data type and object type specified in /2/ (optional) |

In order to access a particular parameter via SDO, the parameter must be addressed with the index and the channel has to be addressed with the sub-index. Devices with only one channel use the sub-indices 0h and 1h only.

To improve the readability of the object dictionary, all objects are allied to a function block model. This model collects all objects belonging to a specific function to a function block. These are the

1. Digital input function block
2. Analogue input function block
3. Digital output function block
4. Analogue output function block
5. Controller function block
6. Alarm function block
7. Device function block

The function block Controller is divided into functions such as

- process values
- admissible value ranges
- basic controller parameters
- parameter set

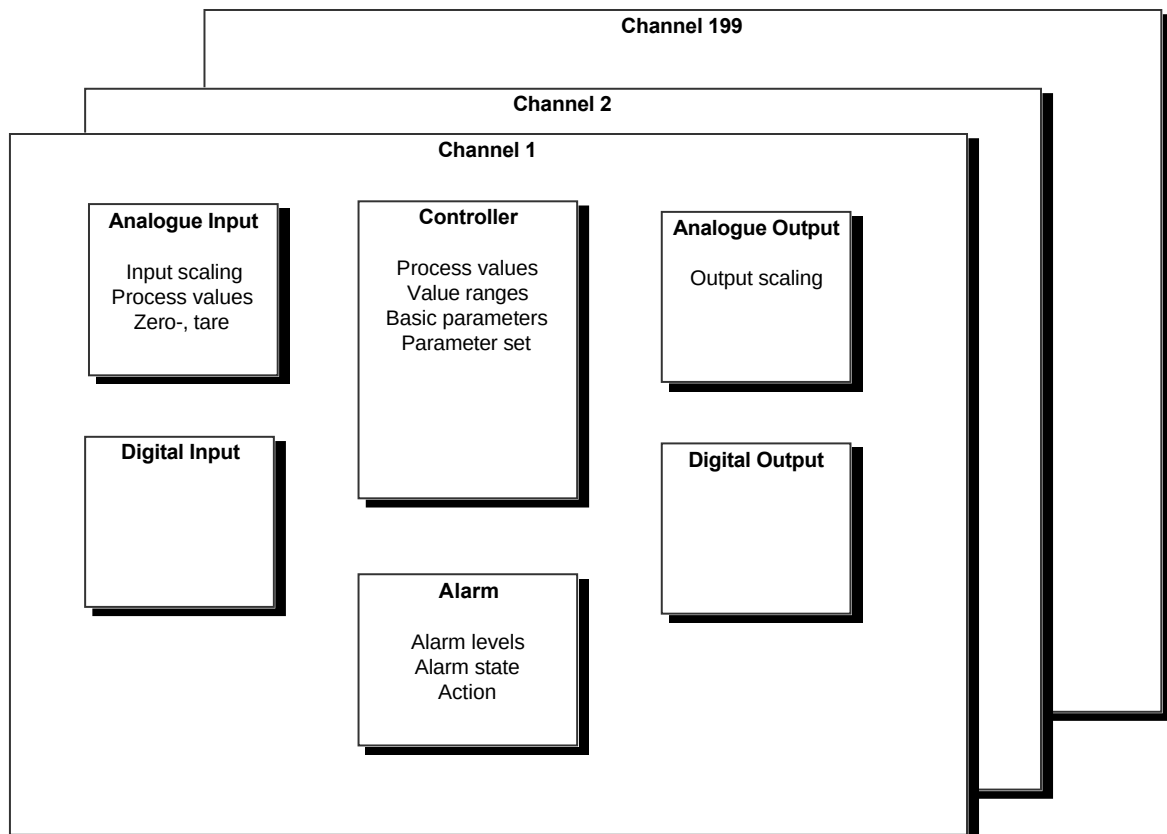


Figure 1: Function block model

#### 4.2.1 Digital input function block

If Module incorporates digital input lines they are addressed according to a subset of the CANopen device profile /3/. Only 1 and 8 Bit access are defined.

#### 4.2.2 Analogue input function block

This functional block may describe the measuring input of a closed loop controller, but may also be used for a sensor with CANopen interface.

The analogue input function block describes how field values are converted to process values.

Examples for **field values** could be:

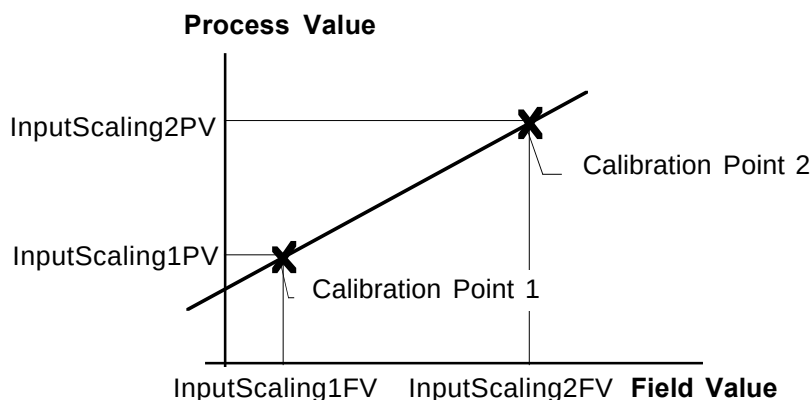
- unscaled readings from analogue digital converter
- counting readings from incremental encoders
- calibrated electrical value at input terminal of measuring transformer (mA, V, mV/V)

These field values are converted to the real physical dimension of the measured quantity, and the result is called "process value".

Examples for **process values** could be:

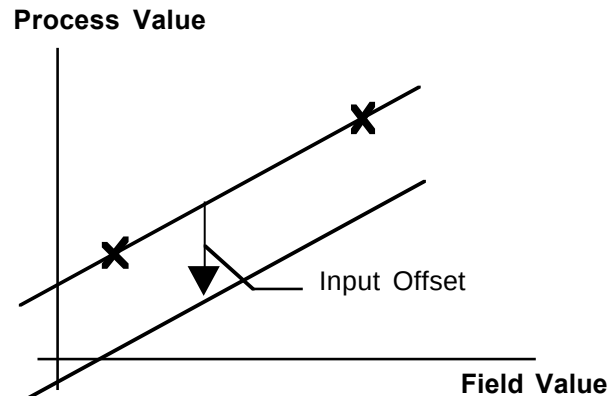
- Value in kg for weight measurement
- Value in Centigrade for temperature measurement
- Value in mm for displacement measurement
- Value in Bar for pressure measurement

The conversion from field value to process value is generally described as a linear transformation. This is defined by two pairs of field value and corresponding process value (Input Scaling 1 FV / Input Scaling 1 PV and Input Scaling 2 FV / Input Scaling 2 PV) called calibration point 1 and 2.



Non-linear transformation (e.g. for thermocouples, PT100 sensors) is Possible and is defined within the parameter "Sensor Type". Then the input scaling values (Input Scaling 1 FV, Input Scaling 1 PV, Input Scaling 2 FV, Input Scaling 2 PV) are meaningless.

The calibration characteristic can be shifted by an additional "**input offset**" value.



Writing "1" to **auto-zero** will set the zero offset value so that the instantaneous measured process value becomes zero.

The **tare-zero** value works like the zero offset value but results in an additional "net process value".

Writing "1" to **auto-tare** will set the tare zero value so that the instantaneous measured net process value will become zero.

The parameters "**Span Start**" and "**Span End**" define the validity range for the process value. If process value exceeds these limits it will be marked as "overflowed".

**Auto-calibration** starts a cycle where the sensor signal is temporarily replaced by a reference signal and internal errors of the measuring components are corrected.

**Test-Mode** defines a static switch of the input lines to internal reference signals for testing purposes.

#### 4.2.3 Digital output function block

If Module incorporates digital output lines they are addressed by a subset of the CANopen device profile according to /3/. Only 1 and 8 Bit access are defined.

#### 4.2.4 Analogue output function block

The analogue output function block describes how process values are converted to analogue output signals (called output field value).

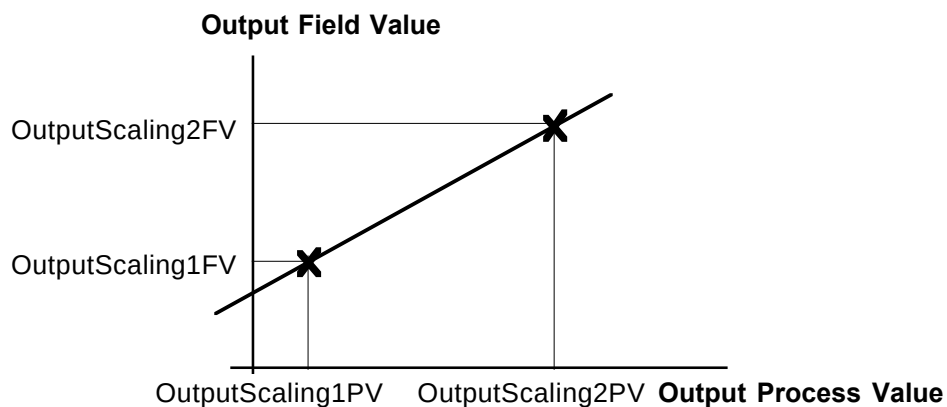
Examples for **output process values** could be:

- Value in kg for weight measurement
- Value in Centigrade for temperature measurement
- Value in mm for displacement measurement
- Value in Bar for pressure measurement

Examples for **output field values** could be:

- unscaled values for digital-analogue converter
- calibrated electrical value for output signal of device (mA, V)

The conversion from output process values to output field values is generally described as a linear transformation. This is defined by two pairs of process values and corresponding output field value (Output Scaling 1 PV / Output Scaling 1 FV and Output Scaling 2 PV / Output Scaling 2 FV).





#### 4.2.5 Controller function block

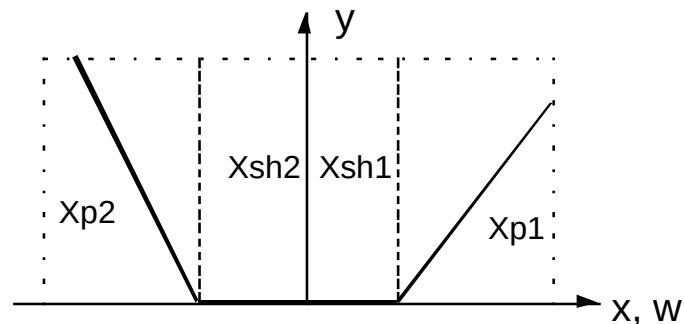
The controller function block describes the PID algorithm for controlling all kinds of process values. The process value might be temperature, pressure, mass flow or any other physical value. The PID controller allows two set-points, a manual override and a self-optimisation.

The controller can be configured as a two-point controller or three-point controller. In the two-point controller mode, the PID algorithm uses the parameters Proportional Band  $Xp1$ , Integral Action Time  $Tn1$ , Derivative Action Time  $Tv1$  and the Cycle Time  $T1$ . In the three-point controller mode the parameters  $Xp2$ ,  $Tn2$ ,  $Tv2$  and  $T2$  will be used, too.

The output of the controller can be limited, e.g. to insure a minimum and a maximum pressure. The output of the controller is a read-only variable. Via the self-optimisation algorithm it is possible to calculate the necessary PID parameters automatically. Because of different implementations by the manufacturer the starting conditions have to be noticed.

The two possible set points ( $W$  and  $W2$ ) can only be set within the limits of the objects „Lower Set Point Limit, ( $W0$ ) and „Upper Set Point Limit, ( $W100$ ). The switching between these set points is performed via the set point switch.

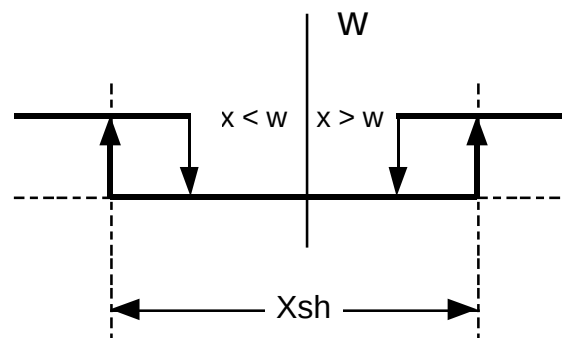
If the controller is configured as three-point controller, a neutral zone can be defined by the parameters  $Xsh1$  and  $Xsh2$ . These parameters have the same physical unit as the current value.



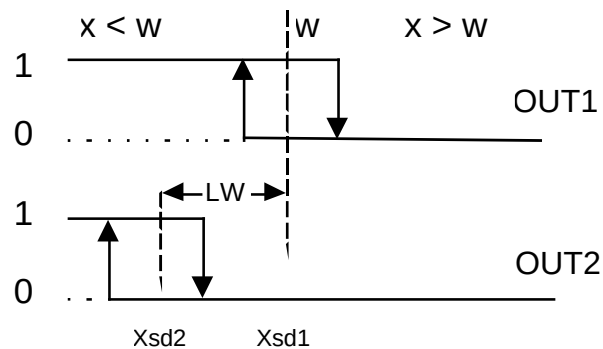
If the controller is configured as three-point stepping controller, a motor response time ( $Tm$ ) has to be defined. The value for  $Xsd$  can be calculated by the equation

$$Xsd = (Tpuls/2) * 0,1 (Xp/Tm).$$

The motor is controlled as shown in the following figure:



The parameters Xsd1, Xsd2 and LW are used to control a signaller with an additional contact. In that case Xsd1 is the switching difference of signaller and Xsd2 is the switching difference of the additional contact. The parameter LW describes the trigger point separation of the additional contact.



#### 4.2.6 Alarm function block

The alarm function block is able to generate alarm messages according to pre-defined alarm conditions like sensor fault, signal value exceeding specified signal levels, difference between two signals exceeding specified level etc. The output of the alarm function block can be mapped to a PDO or may be mapped directly to an output line of the digital output function block. Entering alarm state can also trigger the transmission of emergency messages.

The alarm function block defines objects for up to 16 alarm trigger definitions per channel.

The **Input Value** is compared to **Level** and **Action** is performed when comparison meets the condition specified in **Type**. When for the comparison two input signals are required, the difference between **Input Value** and **Alternate Input Value** is calculated. Input Value and Alternate Input Value may be mapped to a receive PDO or linked to an output of any other function module (analogue input, controller etc.) by **Link** object (e.g. objects 6503h, 6505h). Alarm outputs can be grouped into one of 8 Possible **Groups**, the group alarm is build by a logical or function of the individual alarm states within this group.

Alarms with starting condition are only generated after alarm condition once reaches fault (off) state (after alarm reset). When an alarm **Reset** is performed the state of the alarm is set to off state when input value is within hysteresis and starting condition is reset (relevant only for alarm types with starting condition).

#### 4.2.7 Device function block

This function block includes all objects, which are generally used to control the device.

## 5 ERROR HANDLING

### 5.1 Principle

Emergency Messages are triggered by internal errors in the device and they are assigned the highest Possible priority to ensure that they get access to the bus without latency. By default, the Emergency Messages contain the error field with pre-defined error numbers and additional information.

### 5.2 Error behaviour

If a serious device failure is detected the module shall enter by default autonomously the pre-operational state. If object **1029h** is implemented, the device can be configured to enter alternatively the stopped state or remain in the current state in case of an device failure. Device failures shall include the following communication errors:

- Bus-off conditions of the CAN interface
- Life guarding event with the state 'occurred'
- Heartbeat event with state 'occurred'
- Stop Remote Node indication

Serious device errors also can be caused by device internal failures.

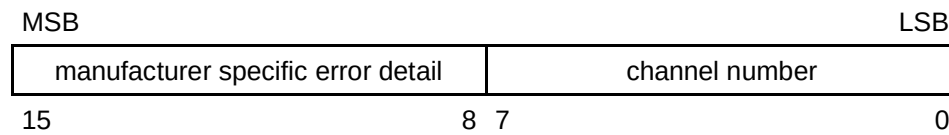
### 5.3 Error code meanings

The error codes are used according to the ones defined within /2/. Within this profile some additional profile specific error codes are defined as follow:

| Error Code | Meaning                                |
|------------|----------------------------------------|
| 2110h      | Short circuit                          |
| 2310h      | Current at outputs too high (Overload) |
| 2320h      | Short circuit at outputs               |
| 2330h      | Load dump at outputs                   |
| 5010h      | Self Test                              |
| 5020h      | Auto-calibration                       |
| 5030h      | Sensor fault                           |
| 5031h      | Fraction of sensor                     |
| 5032h      | Short circuit of sensor                |
| 6310h      | Calibration                            |
| 6311h      | Input Calibration                      |
| 6312h      | Output Calibration                     |
| 6320h      | Controller parameters                  |
| F001h      | Input Overload                         |
| F002h      | Output Overload                        |
| F011h      | Limit Exceeded                         |
| F020h      | Tara Overflow                          |
| F030h      | Self-Optimisation failed               |

#### 5.4 Additional error codes

The additional error code uses the following format



The least significant byte of the additional error code contains the channel number, where the error occurred (for multiple channel devices). The most significant byte of the additional error code can be used manufacturer specific to give more detailed explanation of the error reason.

## 6 PRE-DEFINITIONS

### 6.1 Principle

As a device may include a set of different functional blocks there are more than the four standard PDO's pre-defined. If there are more than 5 PDO's used the corresponding COB-ID needs to be configured.

### 6.2 Naming conventions

The first three characters of CMS name is a device profile identification and is defined for this device profile as: <404>. (According to the number of this standard).

### 6.3 PDO Mapping

The default mapping defines default values for communication objects 1000h to 1FFFh which are not defined by the communication profile (see /2/).

#### 6.3.1 Object 1000h: Device type

The object at index 1000h describes the type of device and its functionality.

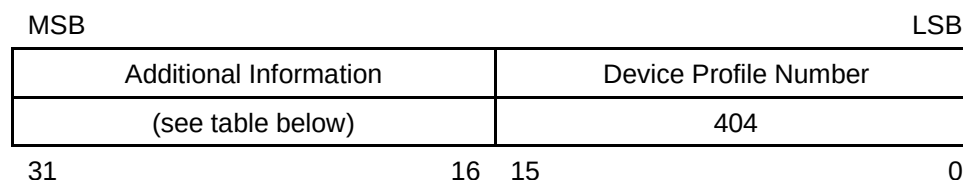


Figure 2: Structure of the Device Type Parameter

Additional information

| Bit | Description           |
|-----|-----------------------|
| 16  | digital input block   |
| 17  | analogue input block  |
| 18  | digital output block  |
| 19  | analogue output block |
| 20  | controller block      |
| 21  | alarm block           |
| 22  | reserved              |
| :   | :                     |
| 31  | reserved              |

Bit description:

- 0 - function block is not present
- 1 - function block is present

Example: Device with analogue input and alarm function has a device type of 0022 0194h.

#### 6.3.2 Object 1001h: Error Register

The device specific bit in the status word is reserved for future use.

### 6.3.3 Object 1029h: Error Behaviour

This object specifies to which state an I/O module shall be set, when a communication error, output error or input error is detected.

#### OBJECT DESCRIPTION

|             |                 |
|-------------|-----------------|
| Index       | 1029h           |
| Name        | Error Behaviour |
| Object Code | ARRAY           |
| Data Type   | Unsigned8       |
| Category    | Optional        |

#### ENTRY DESCRIPTION

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 0h                      |
| Description    | Number of Error Classes |
| Data Type      | Unsigned8               |
| Access         | ro                      |
| Entry Category | Mandatory               |
| PDO Mapping    | No                      |
| Value Range    | 1 – 7h                  |
| Default Value  | No                      |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 1h                  |
| Description    | Communication Error |
| Access         | rw                  |
| Entry Category | Mandatory           |
| PDO Mapping    | No                  |
| Value Range    | see /2/             |
| Default Value  | 0                   |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 2h                  |
| Description    | Digital Input Error |
| Access         | rw                  |
| Entry Category | Optional            |
| PDO Mapping    | No                  |
| Value Range    | see /2/             |
| Default Value  | 0                   |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 3h                   |
| Description    | Analogue Input Error |
| Access         | rw                   |
| Entry Category | Optional             |
| PDO Mapping    | No                   |
| Value Range    | see /2/              |
| Default Value  | 0                    |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 4h                   |
| Description    | Digital Output Error |
| Access         | rw                   |
| Entry Category | Optional             |
| PDO Mapping    | No                   |
| Value Range    | see /2/              |
| Default Value  | 0                    |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 5h                    |
| Description    | Analogue Output Error |
| Access         | rw                    |
| Entry Category | Optional              |
| PDO Mapping    | No                    |
| Value Range    | see /2/               |
| Default Value  | 0                     |

|                |                  |
|----------------|------------------|
| Sub-Index      | 6h               |
| Description    | Controller Error |
| Access         | rw               |
| Entry Category | Optional         |
| PDO Mapping    | No               |
| Value Range    | see /2/          |
| Default Value  | 0                |

|                |             |
|----------------|-------------|
| Sub-Index      | 7h          |
| Description    | Alarm Error |
| Access         | rw          |
| Entry Category | Optional    |
| PDO Mapping    | No          |
| Value Range    | see /2/     |
| Default Value  | 0           |

Note: If this object is not implemented the device shall be set into pre-operational state in the case a communication error is detected.

#### 6.3.4 1st receive PDO parameters (Received Process Values)

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1400h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                  | Default Value |
|-------|-----------|--------------------------|---------------|
| 1600h | 0h        | number of mapped objects | 2             |
|       | 1h        | Received PV 32           | 9F50 0120h    |
|       | 2h        | Received Status          | 6F52 0108h    |

### 6.3.5 1st transmit PDO parameters (Analogue Input)

This PDO is used to transmit input measuring values periodically. By default the transmission of this PDO is initiated by the elapsing of the event timer or as specified by object x133h.

If the transmission type 254 is used, the PDO is transmitted dependent on object x133h. The value of the process value has to change more than the value defined in object x133h between two consecutive transmissions. By default, the signal input is disabled by object 6112h,

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1800h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 1000          |

| Index | Sub-Index | Comment                  | Default Value |
|-------|-----------|--------------------------|---------------|
| 1A00h | 0h        | number of mapped objects | 2 or 3        |
|       | 1h        | AI Input PV 32           | 9130 0120h    |
|       | 2h        | AI Status                | 6150 0108h    |
|       | 3h        | AL 1..8 State *          | 6600 0108h    |

\*) if available

### 6.3.6 2nd receive PDO parameters (Command Values for Controller)

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1401h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |



| Index | Sub-Index | Comment                     | Default Value |
|-------|-----------|-----------------------------|---------------|
| 1601h | 0h        | number of mapped objects    | 3             |
|       | 1h        | CO Set Point W 16           | 7402 0110h    |
|       | 2h        | CO Manual Controller Output | 6412 0110h    |
|       | 3h        | CO Control Byte             | 6425 0108h    |

### 6.3.7 2nd transmit PDO parameters (Effective Controller Output)

This PDO is used to transmit the effective controller output values. The transmission of this PDO is initiated by the elapsing of CO Cycle Time T1 or CO Cycle Time T2.

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1801h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                        | Default Value |
|-------|-----------|--------------------------------|---------------|
| 1A01h | 0h        | number of mapped objects       | 4             |
|       | 1h        | CO Effective Controller Output | 6410 0110h    |
|       | 2h        | CO Effective Current Value 16  | 7400 0110h    |
|       | 3h        | CO Status Information          | 6427 0108h    |
|       | 4h        | AL 1..8 State                  | 6600 0108h    |

### 6.3.8 3rd receive PDO parameters (alternate received process value)

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1402h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                   | Default Value |
|-------|-----------|---------------------------|---------------|
| 1602h | 0h        | number of mapped objects  | 2             |
|       | 1h        | Alternate Received PV 32  | 9F51 0120h    |
|       | 2h        | Alternate Received Status | 6F53 0108h    |

### 6.3.9 3rd transmit PDO parameters (alarm)

By default this PDO is transmitted by a change of the alarm state.

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1802h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                  | Default Value |
|-------|-----------|--------------------------|---------------|
| 1A02h | 0h        | number of mapped objects | 3             |
|       | 1h        | AL 1..8 State            | 6600 0108h    |
|       | 2h        | AL 9..16 State           | 6601 0108h    |
|       | 3h        | AL Group 1..8 State      | 6603 0108h    |

### 6.3.10 4th receive PDO parameters (Digital Output)

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1403h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                       | Default Value |
|-------|-----------|-------------------------------|---------------|
| 1603h | 0h        | number of mapped objects      | 8             |
|       | 1h        | DO Write State 8 Output Lines | 6200 0108h    |
|       | 2h        | DO Write State 8 Output Lines | 6200 0208h    |
|       | 3h        | DO Write State 8 Output Lines | 6200 0308h    |
|       | 4h        | DO Write State 8 Output Lines | 6200 0408h    |
|       | 5h        | DO Write State 8 Output Lines | 6200 0508h    |
|       | 6h        | DO Write State 8 Output Lines | 6200 0608h    |
|       | 7h        | DO Write State 8 Output Lines | 6200 0708h    |
|       | 8h        | DO Write State 8 Output Lines | 6200 0808h    |

#### 6.3.11 4th transmit PDO parameters (Digital Input)

This PDO is transmitted by default when there is any change in the digital input lines.

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1803h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                     | Default Value |
|-------|-----------|-----------------------------|---------------|
| 1A03h | 0h        | number of mapped objects    | 8             |
|       | 1h        | DI Read state 8 Input Lines | 6000 0108h    |
|       | 2h        | DI Read state 8 Input Lines | 6000 0208h    |
|       | 3h        | DI Read state 8 Input Lines | 6000 0308h    |
|       | 4h        | DI Read state 8 Input Lines | 6000 0408h    |
|       | 5h        | DI Read state 8 Input Lines | 6000 0508h    |
|       | 6h        | DI Read state 8 Input Lines | 6000 0608h    |
|       | 7h        | DI Read state 8 Input Lines | 6000 0708h    |
|       | 8h        | DI Read state 8 Input Lines | 6000 0808h    |

#### 6.3.12 5th receive PDO parameters (Control Bytes)

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1404h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                  | Default Value |
|-------|-----------|--------------------------|---------------|
| 1604h | 0h        | number of mapped objects | 2             |
|       | 1h        | AI Control Byte          | 6160 0108h    |
|       | 2h        | CO control byte          | 6425 0108h    |

### 6.3.13 6th receive PDO parameters (Multiplexor PDO)

This PDO is defined according to /2/. It can be used to write to any object in the object dictionary and may be useful in devices with a large number of channels.

| Index | Sub-Index | Comment            | Default Value |
|-------|-----------|--------------------|---------------|
| 1404h | 0h        | number of entries  | see /2/       |
|       | 1h        | COB-ID used by PDO | see /2/       |
|       | 2h        | transmission type  | 255           |
|       | 3h        | inhibit time       | 0             |
|       | 4h        | reserved           |               |
|       | 5h        | event timer        | 0             |

| Index | Sub-Index | Comment                                       | Default Value |
|-------|-----------|-----------------------------------------------|---------------|
| 1604h | 0h        | number of mapped objects<br>(Multiplexor PDO) | 255           |

## 7 OBJECT DICTIONARY

In many applications it is not suitable to stick to a specific data type. In a pure temperature controller application it may be sufficient to transmit the current value as Integer16, whereas in a pressure controller application it may be convenient to use floating point numbers. For this reason the device profile supports four different data types: floating point numbers (Float) as well as 16 bit, 24 bit and 32 bit integers (Integer16, Integer24 and Integer32). To have a better access to the objects in the dictionary, it is divided into four sections:

| Index           | Data Type                      |
|-----------------|--------------------------------|
| 6000h ... 6FFFh | Floating point numbers (Float) |
| 7000h ... 7FFFh | Integer16                      |
| 8000h ... 8FFFh | Integer24                      |
| 9000h ... 9FFFh | Integer32                      |

As mentioned in section 4, the universal device is divided in several function blocks. To give the user an easy access to the dictionary, the following structure is used:

| Index           | Data Type             |
|-----------------|-----------------------|
| x000h ... x0FFh | Digital input block   |
| x100h ... x1FFh | Analogue input block  |
| x200h ... x2FFh | Digital output block  |
| x300h ... x3FFh | Analogue output block |
| x400h ... x4FFh | Controller block      |
| x500h ... x5FFh | Alarm function block  |
| x600h ... xEFFh | reserved              |
| xF00h ... xFFFh | Device function block |

The letter „x“ has to be replaced by the number 6, 7, 8, or 9 for the data type.

## 7.1 Survey object dictionary

### 7.1.1 Digital input function block

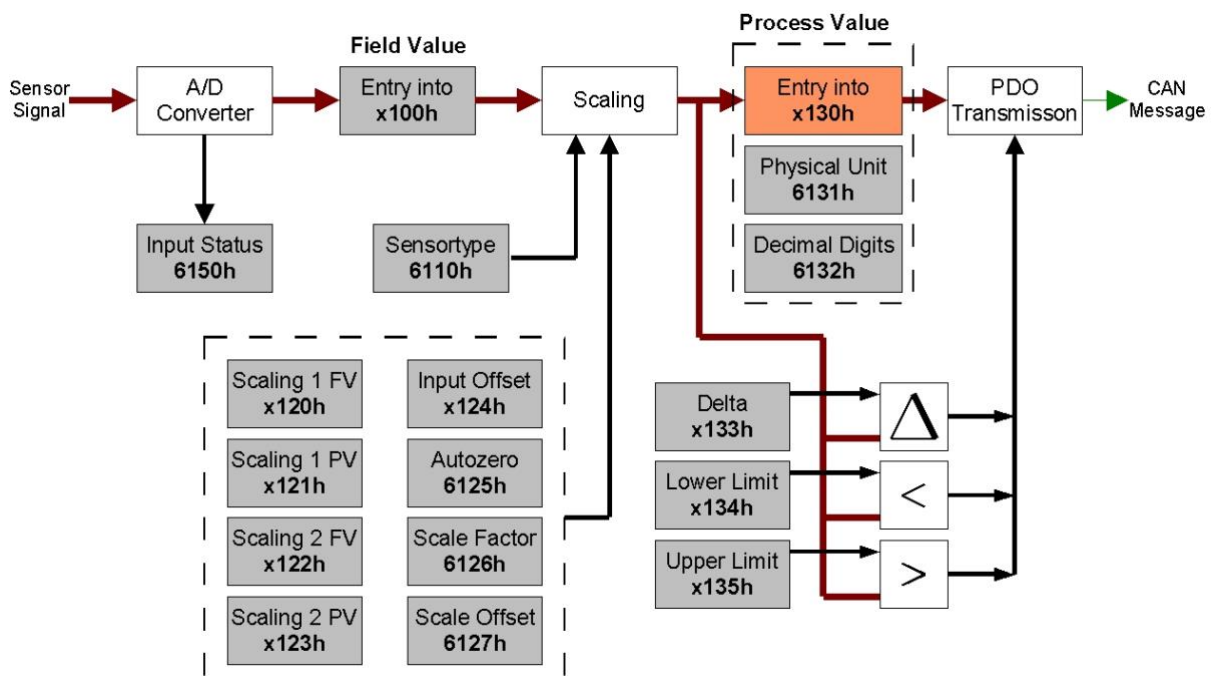
If Module incorporates digital input lines they may be addressed by the CANopen device profile according to device profile specified in /3/. Only 1 and 8 Bit access is defined.

| Index | Object Code | Description                 | Data Type | Acc | PDO mapping | Category |
|-------|-------------|-----------------------------|-----------|-----|-------------|----------|
| 6000h | ARRAY       | DI Read state 8 Input Lines | Unsigned8 | ro  | Possible    | C        |
| 6002h | ARRAY       | DI Polarity 8 Input Lines   | Unsigned8 | rw  | No          | O        |
|       |             |                             |           |     |             |          |
| 6020h | ARRAY       | DI Read state 1 Input Lines | Boolean   | ro  | Possible    | O        |
| 6030h | ARRAY       | DI Polarity 1 Input Lines   | Boolean   | rw  | No          | O        |

### 7.1.2 Analogue input function block

| Index                         | Object Code | Description                 | Data Type      | Acc | PDO mapping | Category |
|-------------------------------|-------------|-----------------------------|----------------|-----|-------------|----------|
| x100h                         | ARRAY       | AI Input FV                 | spec. by index | ro  | Possible    | C/O      |
| 6101h                         | reserved    |                             |                |     |             |          |
| 6102h                         | reserved    |                             |                |     |             |          |
| x103h                         | ARRAY       | AI Interrupt delta input FV | spec. by index | rw  | No          | O        |
| <b>Sensor:</b>                |             |                             |                |     |             |          |
| 6110h                         | ARRAY       | AI Sensor type              | Unsigned16     | rw  | No          | O        |
| 6111h                         | ARRAY       | AI Autocalibration          | Unsigned32     | wo  | No          | O        |
| 6112h                         | ARRAY       | AI Operating mode           | Unsigned8      | rw  | No          | O        |
| 6113h                         | reserved    |                             |                |     |             |          |
| 6114h                         | ARRAY       | AI ADC sample rate          | Unsigned32     | ro  | No          | O        |
| <b>Scaling:</b>               |             |                             |                |     |             |          |
| x120h                         | ARRAY       | AI Input scaling 1 FV       | spec. by index | rw  | No          | O        |
| x121h                         | ARRAY       | AI Input scaling 1 PV       | spec. by index | rw  | No          | O        |
| x122h                         | ARRAY       | AI Input scaling 2 FV       | spec. by index | rw  | No          | O        |
| x123h                         | ARRAY       | AI Input scaling 2 PV       | spec. by index | rw  | No          | O        |
| x124h                         | ARRAY       | AI Input offset             | spec. by index | rw  | No          | O        |
| 6125h                         | ARRAY       | AI Autozero                 | Unsigned32     | wo  | No          | O        |
| 6126h                         | ARRAY       | AI Scaling factor           | Float          | rw  | No          | O        |
| 6127h                         | ARRAY       | A Scaling offset            | Float          | rw  | No          | O        |
| <b>Process value reading:</b> |             |                             |                |     |             |          |
| x130h                         | ARRAY       | AI Input PV                 | spec. by index | ro  | Possible    | C        |
| 6131h                         | ARRAY       | AI Physical unit PV         | Unsigned32     | rw  | No          | C        |
| 6132h                         | ARRAY       | AI Decimal digits PV        | Unsigned8      | rw  | No          | C        |

| Index                                      | Object Code | Description                       | Data Type      | Acc | PDO mapping | Category |
|--------------------------------------------|-------------|-----------------------------------|----------------|-----|-------------|----------|
| x133h                                      | ARRAY       | AI Interrupt delta input PV       | spec. by index | rw  | No          | O        |
| x134h                                      | ARRAY       | AI Interrupt lower limit input PV | spec. by index | rw  | No          | O        |
| x135h                                      | ARRAY       | AI Interrupt upper limit input PV | spec. by index | rw  | No          | O        |
| <b>Tara unit:</b>                          |             |                                   |                |     |             |          |
| x138h                                      | ARRAY       | AI Tare zero                      | spec. by index | rw  | No          | O        |
| 6139h                                      | ARRAY       | AI Autotare                       | Unsigned32     | wo  | No          | O        |
| x140h                                      | ARRAY       | AI Net PV                         | spec. by index | ro  | Possible    | O        |
| x143h                                      | ARRAY       | AI Interrupt delta net PV         | spec. by index | rw  | No          | O        |
| x144h                                      | ARRAY       | AI Interrupt lower limit net PV   | spec. by index | rw  | No          | O        |
| x145h                                      | ARRAY       | AI Interrupt upper limit net PV   | spec. by index | rw  | No          | O        |
| <b>Overflow limits for process values:</b> |             |                                   |                |     |             |          |
| x148h                                      | ARRAY       | AI Span start                     | spec. by index | rw  | No          | O        |
| x149h                                      | ARRAY       | AI Span end                       | spec. by index | rw  | No          | O        |
| <b>Input status:</b>                       |             |                                   |                |     |             |          |
| 6150h                                      | ARRAY       | AI Status                         | Unsigned8      | ro  | Possible    | O        |
| 6160h                                      | ARRAY       | AI Control byte                   | Unsigned8      | wo  | Possible    | O        |
| 6161h                                      | ARRAY       | AI Control byte enable            | Unsigned8      | rw  | No          | O        |
| 61A0h                                      | ARRAY       | AI Filter type                    | Unsigned8      | rw  | No          | O        |
| 61A1h                                      | ARRAY       | AI Filter constant                | Unsigned16     | rw  | No          | O        |



### 7.1.3 Digital output function block

If Module incorporates digital output lines they may be addressed by the CANopen device profile according to device profile specified in /3/. Only 1 and 8 Bit access is defined.

| Index | Object Code | Description                   | Data Type  | Acc | PDO mapping | Category |
|-------|-------------|-------------------------------|------------|-----|-------------|----------|
| 6200h | ARRAY       | DO Write state 8 output lines | Unsigned8  | rw  | Possible    | C        |
| 6202h | ARRAY       | DO Polarity 8 output lines    | Unsigned8  | rw  | No          | O        |
| 6206h | ARRAY       | DO Fault mode 8 output lines  | Unsigned8  | rw  | No          | O        |
| 6207h | ARRAY       | DO Fault state 8 output lines | Unsigned8  | rw  | No          | O        |
| 6210h | RECORD      | DO Link object 8 output lines | PDOMapping | rw  | No          | O        |
|       |             |                               |            |     |             |          |
| 6220h | ARRAY       | DO Write state 1 output lines | Boolean    | rw  | Possible    | O        |
| 6240h | ARRAY       | DO Polarity 1 output lines    | Boolean    | rw  | No          | O        |
| 6250h | ARRAY       | DO Fault mode 1 output lines  | Boolean    | rw  | No          | O        |
| 6260h | ARRAY       | DO Fault state 1 output lines | Boolean    | rw  | No          | O        |
| 6280h | RECORD      | DO Link object 1 output lines | PDOMapping | rw  | No          | O        |

### 7.1.4 Analogue output function block

| Index | Object Code | Description            | Data Type      | Acc | PDO mapping | Category |
|-------|-------------|------------------------|----------------|-----|-------------|----------|
| x300h | ARRAY       | AO Output PV           | spec. by index | rw  | Possible    | O        |
| 6301h | ARRAY       | AO Physical unit PV    | Unsigned32     | rw  | No          | O        |
| 6302h | ARRAY       | AO Decimal digits PV   | Unsigned8      | rw  | No          | O        |
| 6303h | RECORD      | AO Link output PV      | PDOMapping     | rw  | No          | O        |
| 6310h | ARRAY       | AO Output type         | Unsigned16     | rw  | No          | O        |
| x320h | ARRAY       | AO Output scaling 1 PV | spec. by index | rw  | No          | O        |
| x321h | ARRAY       | AO Output scaling 1 FV | spec. by index | rw  | No          | O        |
| x322h | ARRAY       | AO Output scaling 2 PV | spec. by index | rw  | No          | O        |
| x323h | ARRAY       | AO Output scaling 2 FV | spec. by index | rw  | No          | O        |
| x330h | ARRAY       | AO Output FV           | spec. by index | rw  | Possible    | O        |
| 6331h | ARRAY       | AO Physical unit FV    | Unsigned32     | rw  | No          | O        |
| 6332h | ARRAY       | AO Decimal digits FV   | Unsigned8      | rw  | No          | O        |
| 6340h | ARRAY       | AO Fault mode          | Unsigned8      | rw  | No          | O        |
| x341h | ARRAY       | AO Fault FV            | spec. by index | rw  | No          | O        |



## 7.1.5 Controller function block

| Index | Object Code | Description                                 | Data Type      | Acc | PDO mapping | Category |
|-------|-------------|---------------------------------------------|----------------|-----|-------------|----------|
| x400h | ARRAY       | CO Effective current value Xeff             | spec. by index | rw  | Possible    | M        |
| x401h | ARRAY       | CO Effective set point Weff                 | spec. by index | ro  | Possible    | M        |
| x402h | ARRAY       | CO Set point W                              | spec. by index | rw  | Possible    | M        |
| x403h | ARRAY       | CO 2nd set point W2                         | spec. by index | rw  | Possible    | M        |
| x404h | ARRAY       | CO Lower set point limit W0                 | spec. by index | rw  | No          | O        |
| x405h | ARRAY       | CO Upper set point limit W100               | spec. by index | rw  | No          | O        |
| 6406h | ARRAY       | CO Physical unit current value / set point  | Unsigned32     | rw  | No          | M        |
| 6407h | ARRAY       | CO Decimal digits current value / set point | Unsigned8      | rw  | No          | M        |
|       |             |                                             |                |     |             |          |
| 6410h | ARRAY       | CO Effective controller output Y            | Unsigned16     | ro  | Possible    | M        |
| 6411h | ARRAY       | CO Differential controller output Ydiff     | Unsigned16     | ro  | Possible    | O        |
| 6412h | ARRAY       | CO Manual controller output                 | Unsigned16     | rw  | Possible    | O        |
| 6413h | ARRAY       | CO Lower controller output limit Ymin       | Unsigned16     | rw  | No          | O        |
| 6414h | ARRAY       | CO Upper controller output limit Ymax       | Unsigned16     | rw  | No          | O        |
| 6415h | ARRAY       | CO Physical unit controller output          | Unsigned32     | rw  | No          | M        |
| 6416h | reserved    |                                             |                |     |             |          |
|       |             |                                             |                |     |             |          |
| 6420h | ARRAY       | CO Set point switch W/W2                    | Boolean        | rw  | Possible    | M        |
| 6421h | ARRAY       | CO Automatic / manual mode A/M              | Boolean        | rw  | Possible    | M        |
| 6422h | ARRAY       | CO Controller on / off                      | Boolean        | rw  | Possible    | M        |
| 6423h | ARRAY       | CO Controller mode                          | Unsigned8      | rw  | Possible    | M        |
| 6424h | ARRAY       | CO Self optimization on / off               | Boolean        | rw  | Possible    | O        |
| 6425h | ARRAY       | CO control byte                             | Unsigned8      | rw  | Possible    | M        |
| 6426h | ARRAY       | CO control byte enable                      | Unsigned8      | rw  | No          | O        |
| 6427h | ARRAY       | CO status word                              | Unsigned16     | ro  | Possible    | M        |
|       |             |                                             |                |     |             |          |
| x430h | ARRAY       | CO Minimum pulse time Tmin1                 | spec. by index | rw  | No          | O        |
| x431h | ARRAY       | CO Minimum pulse time Tmin2                 | spec. by index | rw  | No          | O        |
| x432h | ARRAY       | CO Motor response time Tm                   | spec. by index | rw  | No          | O        |
| x433h | ARRAY       | CO Min. pulse length Tpuls                  | spec. by index | rw  | No          | O        |
| 6434h | ARRAY       | CO Physical unit pulse timing               | Unsigned32     | rw  | No          | O        |
| 6435h | ARRAY       | CO Decimal digits pulse timing              | Unsigned8      | rw  | No          | O        |
|       |             |                                             |                |     |             |          |
| x440h | ARRAY       | CO Neutral zone 3point Xsh1                 | spec. by index | rw  | No          | O        |

| Index | Object Code | Description                                    | Data Type      | Acc | PDO mapping | Category |
|-------|-------------|------------------------------------------------|----------------|-----|-------------|----------|
| x441h | ARRAY       | CO Neutral zone 3point Xsh2                    | spec. by index | rw  | No          | O        |
| x442h | ARRAY       | CO Neutral zone 3point stepping Xsh            | spec. by index | rw  | No          | O        |
| x443h | ARRAY       | CO Switching difference of signaller Xsd1      | spec. by index | rw  | No          | O        |
| x444h | ARRAY       | CO Switching difference of add. Contact Xsd2   | spec. by index | rw  | No          | O        |
| x445h | ARRAY       | CO Trigger point separation of add. Contact LW | spec. by index | rw  | No          | O        |
|       |             |                                                |                |     |             |          |
| x450h | ARRAY       | CO Proportional band Xp1                       | spec. by index | rw  | No          | O        |
| x451h | ARRAY       | CO Proportional band Xp2                       | spec. by index | rw  | No          | O        |
| x452h | ARRAY       | CO Integral action time Tn1                    | spec. by index | rw  | No          | O        |
| x453h | ARRAY       | CO Integral action time Tn2                    | spec. by index | rw  | No          | O        |
| x454h | ARRAY       | CO Derivative action time Tv1                  | spec. by index | rw  | No          | O        |
| x455h | ARRAY       | CO Derivative action time Tv2                  | spec. by index | rw  | No          | O        |
| x456h | ARRAY       | CO Cycle time T1                               | spec. by index | rw  | No          | O        |
| x457h | ARRAY       | CO Cycle time T2                               | spec. by index | rw  | No          | O        |
| 6458h | ARRAY       | CO Physical unit PID timing                    | Unsigned32     | rw  | No          | O        |
| 6459h | ARRAY       | CO Decimal digits PID timing                   | Unsigned8      | ro  | No          | O        |

## 7.1.6 Alarm function block

| Index | Object Code | Description                 | Data Type      | Acc  | PDO mapping | Category |
|-------|-------------|-----------------------------|----------------|------|-------------|----------|
| x500h | ARRAY       | AL 1 Input value            | spec. by index | rw   | Possible    | M        |
| 6503h | RECORD      | AL 1 Link input             | PDOmapping     | rw   | No          | O        |
| x504h | ARRAY       | AL 1 Alternate input value  | spec. by index | rw   | No          | O        |
| 6505h | RECORD      | AL 1 Link alternate input   | PDOmapping     | rw   | No          | O        |
| 6506h | ARRAY       | AL 1 Fault mode             | Unsigned8      | rw   | No          | O        |
| 6507h | ARRAY       | AL 1 Fault state            | Boolean        | rw   | No          | O        |
| 6508h | ARRAY       | AL 1 Type                   | Unsigned8      | rw   | No          | M        |
| 6509h | ARRAY       | AL 1 Action                 | Unsigned8      | rw   | No          | M        |
| x50Ah | ARRAY       | AL 1 Level                  | spec. by index | rw   | No          | M        |
| x50Bh | ARRAY       | AL 1 Hysteresis             | spec. by index | rw   | No          | O        |
| 650Ch | ARRAY       | AL 1 Group                  | Unsigned8      | rw   | No          | O        |
| 650Dh | ARRAY       | AL 1 State                  | Boolean        | ro   | Possible    | O        |
| 650Eh | ARRAY       | AL 1 Reset                  | Boolean        | wo   | No          | O        |
| ::::  | ::::        | ::::                        | ::::           | :::: | ::::        | ::::     |
| x5F0h | ARRAY       | AL 16 Input value           | spec. by index | rw   | Possible    | M        |
| 65F3h | RECORD      | AL 16 Link input            | PDOmapping     | rw   | No          | O        |
| x5F4h | ARRAY       | AL 16 Alternate input value | spec. by index | rw   | No          | O        |
| 65F5h | RECORD      | AL 16 Link alternate input  | PDOmapping     | rw   | No          | O        |
| 65F6h | ARRAY       | AL 16 Fault mode            | Unsigned8      | rw   | No          | O        |
| 65F7h | ARRAY       | AL 16 Fault state           | Boolean        | rw   | No          | O        |
| 65F8h | ARRAY       | AL 16 Type                  | Unsigned8      | rw   | No          | M        |
| 65F9h | ARRAY       | AL 16 Action                | Unsigned8      | rw   | No          | M        |
| x5FAh | ARRAY       | AL 16 Level                 | spec. by index | rw   | No          | M        |
| x5FBh | ARRAY       | AL 16 Hysteresis            | spec. by index | rw   | No          | O        |
| 65FCh | ARRAY       | AL 16 Group                 | Unsigned8      | rw   | No          | O        |
| 65FDh | ARRAY       | AL 16 State                 | Boolean        | ro   | Possible    | O        |
| 65FEh | ARRAY       | AL 16 Reset                 | Boolean        | wo   | No          | O        |
| 6600h | ARRAY       | AL 1..8 State               | Unsigned8      | ro   | Possible    | M        |
| 6601h | ARRAY       | AL 9..16 State              | Unsigned8      | ro   | Possible    | O        |
| 6602h | VAR         | AL General state            | Boolean        | ro   | Possible    | O        |
| 6603h | VAR         | AL Group 1..8 State         | Unsigned8      | ro   | Possible    | O        |
| 6610h | VAR         | AL General Reset            | Boolean        | wo   | Possible    | O        |
| 6611h | ARRAY       | AL 1..8 Reset               | Unsigned8      | wo   | Possible    | O        |
| 6612h | ARRAY       | AL 9..16 Reset              | Unsigned8      | wo   | Possible    | O        |

- Alarm output lines are configured by using the output function block and linking the lines to the alarm state. Here also polarity etc. can be defined for these output lines.
- Splitting the Alarm State 1..8 and 9..16 into two single byte objects helps keeping PDOs small when Not more than 8 alarms are used.
- Several alarms can be combined by logical „or“ function into group alarms.
- The alarm interrupt will cause asynchronous transmission of 3rd transmit PDO.

#### 7.1.7 Device function block

| Index | Object Code | Description               | Data Type      | Acc | PDO mapping | Category |
|-------|-------------|---------------------------|----------------|-----|-------------|----------|
| 6F20h | ARRAY       | Life counter              | Unsigned8      | ro  | Possible    | O        |
| 6F30h | ARRAY       | Receive PDO check         | Unsigned8      | wo  | Possible    | O        |
| 6F31h | ARRAY       | Transmit PDO check        | Unsigned8      | ro  | Possible    | O        |
| xF50h | ARRAY       | Received PV               | spec. by index | rw  | Possible    | M        |
| xF51h | ARRAY       | Alternate received PV     | spec. by index | rw  | Possible    | O        |
| 6F52h | ARRAY       | Received status           | Unsigned8      | rw  | Possible    | M        |
| 6F53h | ARRAY       | Alternate received status | Unsigned8      | rw  | Possible    | O        |
| 6F60h | VAR         | Transmission enable       | Boolean        | rw  | No          | O        |

## 7.2 Detailed object specification

### 7.2.1 Object 6000h: DI Read state 8 Input Lines

This object shall read groups of 8 input lines as 8-bit information. A maximum of 199 x 8-bit inputs is addressable (1592 inputs). This object is mandatory for modules, which supports digital input lines, and shall support all implemented input lines.

#### OBJECT DESCRIPTION

|             |                                         |
|-------------|-----------------------------------------|
| Index       | 6000h                                   |
| Name        | DI Read state 8 Input Lines             |
| Object Code | ARRAY                                   |
| Data Type   | Unsigned8                               |
| Category    | Conditional: device with digital inputs |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Access         | ro                |
| Entry Category | Mandatory         |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | DI Read state 8 Input Lines 1 |
| Access         | ro                            |
| Entry Category | Mandatory                     |
| PDO Mapping    | Possible                      |
| Value Range    | Unsigned8                     |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | DI Read state 8 Input Lines 2 |
| Access         | ro                            |
| Entry Category | Optional                      |
| PDO Mapping    | Possible                      |
| Value Range    | Unsigned8                     |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | DI Read state 8 Input Lines 199 |
| Access         | ro                              |
| Entry Category | Optional                        |
| PDO Mapping    | Possible                        |
| Value Range    | Unsigned8                       |
| Default Value  | No                              |

### 7.2.2 Object 6002h: DI Polarity 8 Input Lines

This object shall define the polarity of a group of 8 input lines. Input polarity can be inverted individually.

0 = input not inverted

1 = input inverted

If the object is not supported the device shall behave accordingly to the default value.

#### OBJECT DESCRIPTION

|             |                           |
|-------------|---------------------------|
| Index       | 6002h                     |
| Name        | DI Polarity 8 Input Lines |
| Object Code | ARRAY                     |
| Data Type   | Unsigned8                 |
| Category    | Optional                  |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Access         | ro                |
| Entry Category | Mandatory         |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | 1h                          |
| Description    | DI Polarity 8 Input Lines 1 |
| Access         | rw                          |
| Entry Category | Mandatory                   |
| PDO Mapping    | No                          |
| Value Range    | Unsigned8                   |
| Default Value  | 0                           |

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | 2h                          |
| Description    | DI Polarity 8 Input Lines 2 |
| Access         | rw                          |
| Entry Category | Optional                    |
| PDO Mapping    | No                          |
| Value Range    | Unsigned8                   |
| Default Value  | 0                           |

to

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | C7h                           |
| Description    | DI Polarity 8 Input Lines 199 |
| Access         | Rw                            |
| Entry Category | Optional                      |
| PDO Mapping    | No                            |
| Value Range    | Unsigned8                     |
| Default Value  | 0                             |

### 7.2.3 Object 6020h: DI Read state 1 Input Lines

This object shall read single input lines information. A maximum of 199 input lines is addressable at this index.

#### OBJECT DESCRIPTION

|             |                             |
|-------------|-----------------------------|
| Index       | 6020h                       |
| Name        | DI Read state 1 Input Lines |
| Object Code | ARRAY                       |
| Data Type   | Boolean                     |
| Category    | Optional                    |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Access         | ro                |
| Entry Category | Mandatory         |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | DI Read state 1 Input Lines 1 |
| Data Type      | Boolean                       |
| Access         | ro                            |
| Entry Category | Mandatory                     |
| PDO Mapping    | Possible                      |
| Value Range    | Boolean                       |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | DI Read state 1 Input Lines 2 |
| Data Type      | Boolean                       |
| Access         | ro                            |
| Entry Category | Optional                      |
| PDO Mapping    | Possible                      |
| Value Range    | Boolean                       |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | DI Read state 1 Input Lines 199 |
| Data Type      | Boolean                         |
| Access         | ro                              |
| Entry Category | Optional                        |
| PDO Mapping    | Possible                        |
| Value Range    | Boolean                         |
| Default Value  | No                              |

#### 7.2.4 Object 6030h: DI Polarity 1 Input Lines

This object shall define the polarity of single input lines. A maximum of 199 input lines is addressable at this index.

0 = input not inverted

1 = input inverted

If the object is not supported the device shall behave accordingly to the default value.

##### OBJECT DESCRIPTION

|             |                           |
|-------------|---------------------------|
| Index       | 6030h                     |
| Name        | DI Polarity 1 Input Lines |
| Object Code | ARRAY                     |
| Data Type   | Boolean                   |
| Category    | Optional                  |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0                 |
| Description    | number of entries |
| Access         | ro                |
| Entry Category | Mandatory         |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | 1h                          |
| Description    | DI Polarity 1 Input Lines 1 |
| Access         | rw                          |
| Entry Category | Mandatory                   |
| PDO Mapping    | No                          |
| Value Range    | Boolean                     |
| Default Value  | 0                           |

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | 2h                          |
| Description    | DI Polarity 1 Input Lines 2 |
| Access         | rw                          |
| Entry Category | Optional                    |
| PDO Mapping    | No                          |
| Value Range    | Boolean                     |
| Default Value  | 0                           |

to

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | C7h                           |
| Description    | DI Polarity 1 Input Lines 199 |
| Access         | rw                            |
| Entry Category | Optional                      |
| PDO Mapping    | No                            |
| Value Range    | Boolean                       |
| Default Value  | 0                             |



### 7.2.5 Object x100h: AI Input FV

This object represents the converted value of an analogue input module, which is not yet scaled to the physical unit of the quantity being measured. Scaling could be e.g. digits of the analogue digital converter used or Ohms for Pt100 temperature measurement.

The value shall be always left adjusted. The remaining bits at the right side of the LSB shall be set to zero.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x100h                                  |
| Name        | AI Input FV                            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Access         | ro                |
| Entry Category | Mandatory         |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AI Input FV 1      |
| Access         | ro                 |
| Entry Category | Mandatory          |
| PDO Mapping    | Possible           |
| Value Range    | Data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AI Input FV 2      |
| Access         | ro                 |
| Entry Category | Optional           |
| PDO Mapping    | Possible           |
| Value Range    | Data type specific |
| Default Value  | No                 |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AI Input FV 199    |
| Access         | ro                 |
| Entry Category | Optional           |
| PDO Mapping    | Possible           |
| Value Range    | Data type specific |
| Default Value  | No                 |

### 7.2.6 Object x103h: AI Interrupt delta input FV

When Input Field Value is mapped to a transmit PDO and its value has changed delta or more since the last PDO transmission then the PDO is transmitted again. A delta value of 0 means, that this function is disabled.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x103h                                  |
| Name        | AI Interrupt delta input FV            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | AI Interrupt delta input FV 1 |
| Entry Category | Mandatory                     |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | Data type specific            |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | AI Interrupt delta input FV 2 |
| Entry Category | Optional                      |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | Data type specific            |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | AI Interrupt delta input FV 199 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | Data type specific              |
| Default Value  | No                              |

**7.2.7 Object 6110h: AI Sensor type**

Specifies the type of sensor, which is connected to the analogue input.

| Value           | Meaning                                |
|-----------------|----------------------------------------|
| 0               | reserved                               |
| 1               | Thermocouple type J                    |
| 2               | Thermocouple type K                    |
| 3               | Thermocouple type L                    |
| 4               | Thermocouple type N                    |
| 5               | Thermocouple type R                    |
| 6               | Thermocouple type S                    |
| 7               | Thermocouple type T                    |
| 8 ... 29        | reserved (other types of thermocouple) |
| 30              | PT100                                  |
| 31              | PT200                                  |
| 32              | PT500                                  |
| 33              | PT1000                                 |
| 34              | PT5000                                 |
| 35              | IR-Sensor                              |
| 36 .. 39        | reserved                               |
| 40              | Voltage                                |
| 41              | +/- 10 V                               |
| 42              | 0 .. 10 V                              |
| 43              | +/- 1 V                                |
| 44              | 0 .. 1 V                               |
| 45              | +/- 100 mV                             |
| 46              | 0 .. 100 mV                            |
| 47 .. 49        | reserved                               |
| 50              | Current                                |
| 51              | 4..20mA                                |
| 52              | 0..20mA                                |
| 53 .. 59        | reserved                               |
| 60              | Frequency                              |
| 61 ... 69       | reserved                               |
| 70              | Strain gauge                           |
| 71              | Strain gauge full bridge               |
| 72              | Strain gauge half bridge               |
| 73              | Strain gauge quarter bridge            |
| 74 .. 79        | reserved                               |
| 80              | LVDT                                   |
| 81 .. 89        | reserved                               |
| 90              | Pressure Transducer                    |
| 91 .. 99        | reserved                               |
| 100             | Temperature transducer                 |
| 101 .. 119      | reserved                               |
| 120             | Potentiometer                          |
| 121 ... 9999    | reserved                               |
| 10000 ... 65535 | manufacturer specific use              |

## OBJECT DESCRIPTION

|             |                |
|-------------|----------------|
| Index       | 6110h          |
| Name        | AI Sensor type |
| Object Code | ARRAY          |
| Data Type   | Unsigned16     |
| Category    | Optional       |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                  |
|----------------|------------------|
| Sub-Index      | 1h               |
| Description    | AI Sensor type 1 |
| Entry Category | Mandatory        |
| Access         | rw               |
| PDO Mapping    | No               |
| Value Range    | Unsigned16       |
| Default Value  | No               |

|                |                  |
|----------------|------------------|
| Sub-Index      | 2h               |
| Description    | AI Sensor type 2 |
| Entry Category | Optional         |
| Access         | rw               |
| PDO Mapping    | No               |
| Value Range    | Unsigned16       |
| Default Value  | No               |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AI Sensor type 199 |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | Unsigned16         |
| Default Value  | No                 |

**7.2.8 Object 6111h: AI Autocalibration**

Writing a signature value of „cali“ to this object starts auto-calibration cycle of the analogue input channel for one time.

If a wrong signature is written, the device refuses to calibrate and respond with an abort message (abort code 0800 002xh)

|           |     |   |   |     |
|-----------|-----|---|---|-----|
| Signature | MSB |   |   | LSB |
| ASCII     | i   | l | a | c   |

hex

69

6C

61

63

## OBJECT DESCRIPTION

|             |                    |
|-------------|--------------------|
| Index       | 6111h              |
| Name        | AI Autocalibration |
| Object Code | ARRAY              |
| Data Type   | Unsigned32         |
| Category    | Optional           |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 1h                   |
| Description    | AI Autocalibration 1 |
| Entry Category | Mandatory            |
| Access         | wo                   |
| PDO Mapping    | No                   |
| Value Range    | Unsigned32           |
| Default Value  | No                   |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 2h                   |
| Description    | AI Autocalibration 2 |
| Entry Category | Optional             |
| Access         | wo                   |
| PDO Mapping    | No                   |
| Value Range    | Unsigned32           |
| Default Value  | No                   |

to

|                |                        |
|----------------|------------------------|
| Sub-Index      | C7h                    |
| Description    | AI Autocalibration 199 |
| Entry Category | Optional               |
| Access         | wo                     |
| PDO Mapping    | No                     |
| Value Range    | Unsigned32             |
| Default Value  | No                     |

### 7.2.9 Object 6112h: AI Operating mode

A value different from 0 sets the analogue input channel to special operating modes.

| Value      | Meaning                     |
|------------|-----------------------------|
| 0          | Channel off (Not operating) |
| 1          | Normal operation            |
| 2 ... 9    | Reserved                    |
| 10 ... 255 | Implementation specific     |

#### OBJECT DESCRIPTION

|             |                   |
|-------------|-------------------|
| Index       | 6112h             |
| Name        | AI Operating mode |
| Object Code | ARRAY             |
| Data Type   | Unsigned8         |
| Category    | Optional          |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0                 |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 1h                  |
| Description    | AI Operating mode 1 |
| Entry Category | Mandatory           |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | Unsigned8           |
| Default Value  | 0                   |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 2h                  |
| Description    | AI Operating mode 2 |
| Entry Category | Optional            |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | Unsigned8           |
| Default Value  | 0                   |

to

|                |                       |
|----------------|-----------------------|
| Sub-Index      | C7h                   |
| Description    | AI Operating mode 199 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned8             |
| Default Value  | 0                     |

### 7.2.10 Object 6114h: AI ADC sample rate

This value has to be the conversion rate used by the AD converter. The value is given in multiples of microseconds.

#### OBJECT DESCRIPTION

|             |                    |
|-------------|--------------------|
| Index       | 6114h              |
| Name        | AI ADC sample rate |
| Object Code | ARRAY              |
| Data Type   | Unsigned32         |
| Category    | Optional           |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0                 |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – C7h           |
| Default Value  | No                |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 1h                   |
| Description    | AI ADC sample rate 1 |
| Entry Category | Mandatory            |
| Access         | ro                   |
| PDO Mapping    | No                   |
| Value Range    | Unsigned32           |
| Default Value  | 0                    |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 2h                   |
| Description    | AI ADC sample rate 2 |
| Entry Category | Optional             |
| Access         | ro                   |
| PDO Mapping    | No                   |
| Value Range    | Unsigned32           |
| Default Value  | 0                    |

to

|                |                        |
|----------------|------------------------|
| Sub-Index      | C7h                    |
| Description    | AI ADC sample rate 199 |
| Entry Category | Optional               |
| Access         | ro                     |
| PDO Mapping    | No                     |
| Value Range    | Unsigned32             |
| Default Value  | 0                      |

#### 7.2.11 Object x120h: AI Input scaling 1 FV

This object defines the field value of the first calibration point for the analogue input channel. It is scaled in physical unit of field value.

##### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x120h                                  |
| Name        | AI Input scaling 1 FV                  |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – 254           |
| Default Value  | No                |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | AI Input scaling 1 FV 1 |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | AI Input scaling 1 FV 2 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

to



|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | AI Input scaling 1 FV 199 |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | data type specific        |
| Default Value  | No                        |

### 7.2.12 Object x121h: AI Input scaling 1 PV

This object defines the process value of the first calibration point for the analogue input channel. It is scaled in physical unit of process value.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x121h                                  |
| Name        | AI Input scaling 1 PV                  |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – 254           |
| Default Value  | No                |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | AI Input scaling 1 PV 1 |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | AI Input scaling 1 PV 2 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

to

|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | AI Input scaling 1 PV 199 |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | data type specific        |
| Default Value  | No                        |

### 7.2.13 Object x122h: AI Input scaling 2 FV

This object defines the field value of the second calibration point for the analogue input channel. It is scaled in physical unit of field value.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x122h                                  |
| Name        | AI Input scaling 2 FV                  |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0                 |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | AI Input scaling 2 FV 1 |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | AI Input scaling 2 FV 2 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

to

|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | AI Input scaling 2 FV 199 |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | data type specific        |
| Default Value  | No                        |

#### 7.2.14 Object x123h: AI Input scaling 2 PV

This object defines the process value of the second calibration point for the analogue input channel. It is scaled in physical unit of process value.

##### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x123h                                  |
| Name        | AI Input scaling 2 PV                  |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | AI Input scaling 2 PV 1 |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | AI Input scaling 2 PV 2 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

to

|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | AI Input scaling 2 PV 199 |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | data type specific        |
| Default Value  | No                        |

### 7.2.15 Object x124h: AI Input offset

This object defines the additional offset value for the analogue input channel. It is scaled in physical unit of process value.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x124h                                  |
| Name        | AI Input offset                        |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AI Input offset 1  |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AI Input offset 2  |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

to

|                |                     |
|----------------|---------------------|
| Sub-Index      | C7h                 |
| Description    | AI Input offset 199 |
| Entry Category | Optional            |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | data type specific  |
| Default Value  | 0                   |

### 7.2.16 Object 6125h: AI Autozero

Writing a signature value of „zero“ to this object causes a modification of the AI Input Offset in such a way that the actual AI Input PV becomes zero. This cycle is performed for one time.

|           |     |    |     |    |
|-----------|-----|----|-----|----|
| Signature | MSB |    | LSB |    |
| ASCII     | o   | r  | e   | z  |
| hex       | 6f  | 72 | 65  | 7A |

#### OBJECT DESCRIPTION

|             |             |
|-------------|-------------|
| Index       | 6125h       |
| Name        | AI Autozero |
| Object Code | ARRAY       |
| Data Type   | Unsigned32  |
| Category    | Optional    |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |               |
|----------------|---------------|
| Sub-Index      | 1h            |
| Description    | AI Autozero 1 |
| Entry Category | Mandatory     |
| Access         | wo            |
| PDO Mapping    | No            |
| Value Range    | Unsigned32    |
| Default Value  | No            |

|                |               |
|----------------|---------------|
| Sub-Index      | 2h            |
| Description    | AI Autozero 2 |
| Entry Category | Optional      |
| Access         | wo            |
| PDO Mapping    | No            |
| Value Range    | Unsigned32    |
| Default Value  | No            |

to

|                |                 |
|----------------|-----------------|
| Sub-Index      | C7h             |
| Description    | AI Autozero 199 |
| Entry Category | Optional        |
| Access         | w0              |
| PDO Mapping    | No              |
| Value Range    | Unsigned32      |
| Default Value  | No              |

### 7.2.17 Object x126h: AI Scaling Factor

This object represents the scaling factor by which the field value needs to be multiplied to get process values.

#### OBJECT DESCRIPTION

|             |                   |
|-------------|-------------------|
| Index       | 6126h             |
| Name        | AI Scaling Factor |
| Object Code | ARRAY             |
| Data Type   | Float             |
| Category    | Optional          |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 1h                  |
| Description    | AI Scaling Factor 1 |
| Entry Category | Mandatory           |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | Float               |
| Default Value  | 1                   |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 2h                  |
| Description    | AI Scaling Factor 2 |
| Entry Category | Optional            |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | Float               |
| Default Value  | 1                   |

to

|                |                       |
|----------------|-----------------------|
| Sub-Index      | C7h                   |
| Description    | AI Scaling Factor 199 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Float                 |
| Default Value  | 1                     |

### 7.2.18 Object 6127h: AI Scaling Offset

This object represents the scaling offset, which is needed to calculate the process values from the field values:

$$\text{Process Value} = (\text{Field Value} * \text{Scaling Factor}) + \text{Scaling Offset}$$

#### OBJECT DESCRIPTION

|                |                   |
|----------------|-------------------|
| Index          | 6127h             |
| Name           | AI Scaling Offset |
| Object Code    | ARRAY             |
| Data Type      | Float             |
| Entry Category | Optional          |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Access         | ro                |
| Entry Category | Mandatory         |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 1h                  |
| Description    | AI Scaling Offset 1 |
| Access         | rw                  |
| Entry Category | Mandatory           |
| PDO Mapping    | No                  |
| Value Range    | Float               |
| Default Value  | 0                   |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 2h                  |
| Description    | AI Scaling Offset 2 |
| Access         | rw                  |
| Entry Category | Optional            |
| PDO Mapping    | No                  |
| Value Range    | Float               |
| Default Value  | 0                   |

to

|                |                       |
|----------------|-----------------------|
| Sub-Index      | C7h                   |
| Description    | AI Scaling Offset 199 |
| Access         | rw                    |
| Entry Category | Optional              |
| PDO Mapping    | No                    |
| Value Range    | Float                 |
| Default Value  | 0                     |

### 7.2.19 Object x130h: AI Input PV

This object represents the result of the input scaling block and gives the measured quantity scaled in the physical unit of process values (e.g. degrees centigrade, kg, kN, mm etc.).

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x130h                                  |
| Name        | AI Input PV                            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Conditional                            |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AI Input PV 1      |
| Entry Category | Mandatory          |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AI Input PV 2      |
| Entry Category | Optional           |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

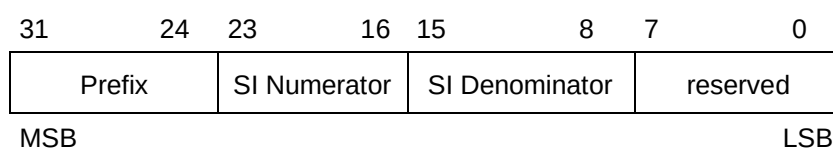
to



|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AI Input PV 199    |
| Entry Category | Optional           |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

### 7.2.20 Object 6131h: AI Physical unit PV

This object shall assign SI units and prefixes for the process values within the analogue input function block. The structure of the SI unit entry is as follows:



The coding of the physical units and prefixes is done according to the /5/. Within this profile, there are some additional physical units specified:

| Code            | Physical Unit      |
|-----------------|--------------------|
| 55 <sub>h</sub> | m/s                |
| 56 <sub>h</sub> | Nm                 |
| A1 <sub>h</sub> | at                 |
| A2 <sub>h</sub> | mmH <sub>2</sub> O |
| A3 <sub>h</sub> | mHG                |
| A4 <sub>h</sub> | atm                |
| AB <sub>h</sub> | PSI                |
| AC <sub>h</sub> | °F                 |

#### OBJECT DESCRIPTION

|             |                     |
|-------------|---------------------|
| Index       | 6131h               |
| Name        | AI Physical unit PV |
| Object Code | ARRAY               |
| Data Type   | Unsigned32          |
| Category    | Conditional         |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 1h                    |
| Description    | AI Physical unit PV 1 |
| Entry Category | Mandatory             |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned32            |
| Default Value  | No                    |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 2h                    |
| Description    | AI Physical unit PV 2 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned32            |
| Default Value  | No                    |

to

|                |                         |
|----------------|-------------------------|
| Sub-Index      | C7h                     |
| Description    | AI Physical unit PV 199 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | Unsigned32              |
| Default Value  | No                      |

#### 7.2.21 Object 6132h: AI Decimal digits PV

This object describes the number of decimal digits following the decimal point for interpretation of data types Integer8, Integer16 and Integer32.

Example: A field value of 1.230 (Float) will be coded as 123 in Integer16 format if number of decimal digits is set to 2.

##### OBJECT DESCRIPTION

|             |                      |
|-------------|----------------------|
| Index       | 6132h                |
| Name        | AI Decimal digits PV |
| Object Code | ARRAY                |
| Data Type   | Unsigned8            |
| Category    | Conditional          |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0                 |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | AI Decimal digits PV 1 |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 9h                 |
| Default Value  | No                     |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | AI Decimal digits PV 2 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 9h                 |
| Default Value  | No                     |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | AI Decimal digits PV 199 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 – 9h                   |
| Default Value  | No                       |

#### 7.2.22 Object x133h: AI Interrupt delta input PV

When Input Process Value is mapped to a transmit PDO and its value has changed delta or more since the last PDO transmission then the PDO is transmitted again. A delta value of 0 means, that this function is disabled.

##### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x133h                                  |
| Name        | AI Interrupt delta input PV            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | AI Interrupt delta input PV 1 |
| Entry Category | Mandatory                     |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | Data type specific            |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | AI Interrupt delta input PV 2 |
| Entry Category | Optional                      |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | Data type specific            |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | AI Interrupt delta input PV 199 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | data type specific              |
| Default Value  | No                              |

### 7.2.23 Object x134h: AI Interrupt lower limit input PV

This object sets the lower limits for interrupt-enabled analogue inputs. The value shall be always left adjusted.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x134h                                  |
| Name        | AI Interrupt lower limit input PV      |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – C7h           |
| Default Value  | No                |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 1h                                  |
| Description    | AI Interrupt lower limit input PV 1 |
| Entry Category | Mandatory                           |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | Data type specific                  |
| Default Value  | No                                  |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 2h                                  |
| Description    | AI Interrupt lower limit input PV 2 |
| Entry Category | Optional                            |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | Data type specific                  |
| Default Value  | No                                  |

to

|                |                                       |
|----------------|---------------------------------------|
| Sub-Index      | C7h                                   |
| Description    | AI Interrupt lower limit input PV 199 |
| Entry Category | Optional                              |
| Access         | rw                                    |
| PDO Mapping    | No                                    |
| Value Range    | data type specific                    |
| Default Value  | No                                    |

#### 7.2.24 Object x135h: AI Interrupt upper limit input PV

This object sets the upper limits for interrupt-enabled analogue inputs. The value shall be always left adjusted.

##### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x135h                                  |
| Name        | AI Interrupt upper limit input PV      |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 1h                                  |
| Description    | AI Interrupt upper limit input PV 1 |
| Entry Category | Mandatory                           |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | data type specific                  |
| Default Value  | No                                  |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 2h                                  |
| Description    | AI Interrupt upper limit input PV 2 |
| Entry Category | Optional                            |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | data type specific                  |
| Default Value  | No                                  |

to

|                |                                       |
|----------------|---------------------------------------|
| Sub-Index      | C7h                                   |
| Description    | AI Interrupt upper limit input PV 199 |
| Entry Category | Optional                              |
| Access         | rw                                    |
| PDO Mapping    | No                                    |
| Value Range    | data type specific                    |
| Default Value  | No                                    |

#### 7.2.25 Object x138h: AI Tare zero

This object defines the tare-offset value. Net process value is calculated by subtracting the tare-offset value from the process value. It is scaled in physical unit of process value.

##### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x138h                                  |
| Name        | AI Tare zero                           |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AI Tare zero 1     |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AI Tare zero 2     |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AI Tare zero 199   |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

### 7.2.26 Object 6139h: AI Autotare

Writing a signature value of „tara“ to this object causes a modification of the AI Tare Zero in such a way that the actual AI Net PV becomes zero. This cycle is processed one time.

|           |     |    |    |    |     |  |  |  |
|-----------|-----|----|----|----|-----|--|--|--|
| Signature | MSB |    |    |    | LSB |  |  |  |
| ASCII     | a   | r  | a  | t  |     |  |  |  |
| hex       | 61  | 72 | 61 | 74 |     |  |  |  |

#### OBJECT DESCRIPTION

|             |             |
|-------------|-------------|
| Index       | 6139h       |
| Name        | AI Autotare |
| Object Code | ARRAY       |
| Data Type   | Unsigned32  |
| Category    | Optional    |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |               |
|----------------|---------------|
| Sub-Index      | 1h            |
| Description    | AI Autotare 1 |
| Entry Category | Mandatory     |
| Access         | wo            |
| PDO Mapping    | No            |
| Value Range    | Unsigned32    |
| Default Value  | No            |

|                |               |
|----------------|---------------|
| Sub-Index      | 2h            |
| Description    | AI Autotare 2 |
| Entry Category | Optional      |
| Access         | wo            |
| PDO Mapping    | No            |
| Value Range    | Unsigned32    |
| Default Value  | No            |

to

|                |                 |
|----------------|-----------------|
| Sub-Index      | C7h             |
| Description    | AI Autotare 199 |
| Entry Category | Optional        |
| Access         | wo              |
| PDO Mapping    | No              |
| Value Range    | Unsigned32      |
| Default Value  | No              |

### 7.2.27 Object x140h: AI Net PV

This read only object represents the result of the tare function block and gives the net measured quantity scaled in the physical unit of process values (e.g. degrees centigrade, kg, kN, mm etc.).

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x140h                                  |
| Name        | AI Net PV                              |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |



|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AI Net PV 1        |
| Entry Category | Mandatory          |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AI Net PV 2        |
| Entry Category | Optional           |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AI Net PV 199      |
| Entry Category | Optional           |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

### 7.2.28 Object x143h: AI Interrupt delta net PV

When Net Process Value is mapped to a transmit PDO and its value has changed delta or more since the last PDO transmission then the PDO is transmitted again. A delta value of 0 means, that this function is disabled.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x143h                                  |
| Name        | AI Interrupt delta net PV              |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | 1h                          |
| Description    | AI Interrupt delta net PV 1 |
| Entry Category | Mandatory                   |
| Access         | rw                          |
| PDO Mapping    | No                          |
| Value Range    | data type specific          |
| Default Value  | No                          |

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | 2h                          |
| Description    | AI Interrupt delta net PV 2 |
| Entry Category | Optional                    |
| Access         | rw                          |
| PDO Mapping    | No                          |
| Value Range    | data type specific          |
| Default Value  | No                          |

to

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | C7h                           |
| Description    | AI Interrupt delta net PV 199 |
| Entry Category | Optional                      |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

### 7.2.29 Object x144h: AI Interrupt lower limit net PV

This object sets the lower limits for interrupt-enabled analogue net inputs. The value shall be always left adjusted.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x144h                                  |
| Name        | AI Interrupt lower limit net PV        |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | 1h                                |
| Description    | AI Interrupt lower limit net PV 1 |
| Entry Category | Mandatory                         |
| Access         | rw                                |
| PDO Mapping    | No                                |
| Value Range    | data type specific                |
| Default Value  | No                                |

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | 2h                                |
| Description    | AI Interrupt lower limit net PV 2 |
| Entry Category | Optional                          |
| Access         | rw                                |
| PDO Mapping    | No                                |
| Value Range    | data type specific                |
| Default Value  | No                                |

to

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | C7h                                 |
| Description    | AI Interrupt lower limit net PV 199 |
| Entry Category | Optional                            |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | data type specific                  |
| Default Value  | No                                  |

### 7.2.30 Object x145h: AI Interrupt upper limit net PV

This object sets the upper limits for interrupt-enabled analogue net inputs. The value shall be always left adjusted.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x145h                                  |
| Name        | AI Interrupt upper limit net PV        |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | 1h                                |
| Description    | AI Interrupt upper limit net PV 1 |
| Entry Category | Mandatory                         |
| Access         | rw                                |
| PDO Mapping    | No                                |
| Value Range    | data type specific                |
| Default Value  | No                                |

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | 2h                                |
| Description    | AI Interrupt upper limit net PV 2 |
| Entry Category | Optional                          |
| Access         | rw                                |
| PDO Mapping    | No                                |
| Value Range    | data type specific                |
| Default Value  | No                                |

to

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | C7h                                 |
| Description    | AI Interrupt upper limit net PV 199 |
| Entry Category | Optional                            |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | data type specific                  |
| Default Value  | No                                  |

### 7.2.31 Object x148h: AI Span start

This value specifies the lower limit where process values are expected. Process values, which are lower than this limit, are marked as negative overloaded.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x148h                                  |
| Name        | <b>AI Span start</b>                   |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | <b>AI Span start 1</b> |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | data type specific     |
| Default Value  | No                     |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | <b>AI Span start 2</b> |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | data type specific     |
| Default Value  | No                     |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | <b>AI Span start 199</b> |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

### 7.2.32 Object x149h: AI Span end

This value specifies the upper limit where process values are expected. Process values exceeding this limit are marked as positive overloaded.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x149h                                  |
| Name        | <b>AI Span end</b>                     |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 1h                   |
| Description    | <b>AI Span end 1</b> |
| Entry Category | Mandatory            |
| Access         | rw                   |
| PDO Mapping    | No                   |
| Value Range    | data type specific   |
| Default Value  | No                   |

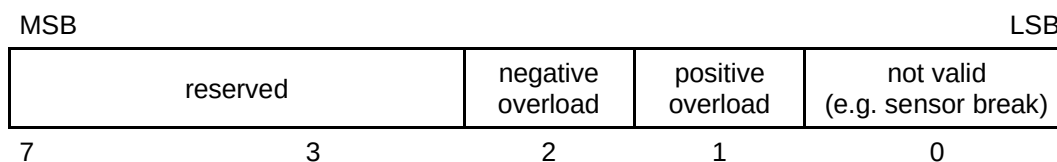
|                |                      |
|----------------|----------------------|
| Sub-Index      | 2h                   |
| Description    | <b>AI Span end 2</b> |
| Entry Category | Optional             |
| Access         | rw                   |
| PDO Mapping    | No                   |
| Value Range    | data type specific   |
| Default Value  | No                   |

to

|                |                        |
|----------------|------------------------|
| Sub-Index      | C7h                    |
| Description    | <b>AI Span end 199</b> |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | data type specific     |
| Default Value  | No                     |

### 7.2.33 Object 6150h: AI Status

This read-only object reflects the status of the analogue input channels.



The combination of bit 1 and bit 2 has not to be possible.

For example:

| Value | Meaning                                     |
|-------|---------------------------------------------|
| 00h   | No failure at analogue input                |
| 03h   | Positive overload, eg.g. Pt100 sensor break |

## OBJECT DESCRIPTION

|             |           |
|-------------|-----------|
| Index       | 6150h     |
| Name        | AI Status |
| Object Code | ARRAY     |
| Data Type   | Unsigned8 |
| Category    | Optional  |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |             |
|----------------|-------------|
| Sub-Index      | 1h          |
| Description    | AI Status 1 |
| Entry Category | Mandatory   |
| Access         | ro          |
| PDO Mapping    | Possible    |
| Value Range    | Unsigned8   |
| Default Value  | No          |

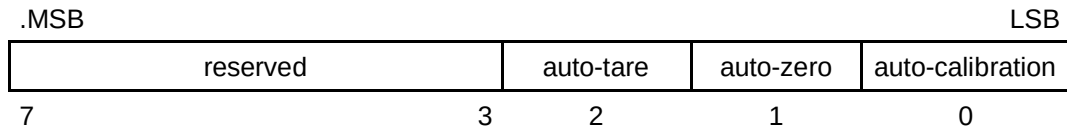
|                |             |
|----------------|-------------|
| Sub-Index      | 2h          |
| Description    | AI Status 2 |
| Entry Category | Optional    |
| Access         | ro          |
| PDO Mapping    | Possible    |
| Value Range    | Unsigned8   |
| Default Value  | No          |

to

|                |               |
|----------------|---------------|
| Sub-Index      | C7h           |
| Description    | AI Status 199 |
| Entry Category | Optional      |
| Access         | ro            |
| PDO Mapping    | Possible      |
| Value Range    | Unsigned8     |
| Default Value  | No            |

**7.2.34 Object 6160h: AI Control byte**

This write-only object starts several control functions of the analogue input function block



For example:

| Value | Meaning                                           |
|-------|---------------------------------------------------|
| 00h   | Access to the control byte object is not possible |
| 03h   | Allow auto-calibration and auto-zero              |

**OBJECT DESCRIPTION**

|             |                        |
|-------------|------------------------|
| Index       | 6160h                  |
| Name        | <b>AI Control byte</b> |
| Object Code | ARRAY                  |
| Data Type   | Unsigned8              |
| Category    | Optional               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | <b>AI Control byte 1</b> |
| Entry Category | Mandatory                |
| Access         | wo                       |
| PDO Mapping    | Possible                 |
| Value Range    | Unsigned8                |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | <b>AI Control byte 2</b> |
| Entry Category | Optional                 |
| Access         | wo                       |
| PDO Mapping    | Possible                 |
| Value Range    | Unsigned8                |
| Default Value  | No                       |

to



|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | <b>AI Control byte 199</b> |
| Entry Category | Optional                   |
| Access         | w0                         |
| PDO Mapping    | Possible                   |
| Value Range    | Unsigned8                  |
| Default Value  | No                         |

### 7.2.35 Object 6161h: AI Control byte enable

Bit mask to enable the single bits in object 6160h (AI Control byte enable).

0 – disabled function

1 – enabled function

#### OBJECT DESCRIPTION

|             |                        |
|-------------|------------------------|
| Index       | 6161h                  |
| Name        | AI Control byte enable |
| Object Code | ARRAY                  |
| Data Type   | Unsigned8              |
| Category    | Optional               |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | AI Control byte enable 1 |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | Unsigned8                |
| Default Value  | 0                        |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | AI Control byte enable 2 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | Unsigned8                |
| Default Value  | 0                        |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | AI Control byte enable 199 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | Unsigned8                  |
| Default Value  | 0                          |

### 7.2.36 Object 61A0h: AI Filter type

The object shall define the type of filter to be used for calculation.

| Value      | Description           |
|------------|-----------------------|
| 0          | No filter             |
| 1          | Moving average        |
| 2          | Repeating average     |
| 3 .. 99    | reserved              |
| 100 .. 255 | Manufacturer specific |

- Calculation of moving average**

There has to be calculated the difference between the current value derived from the A/D converter and the previous calculated average value, This current calculated difference has to be divided by a filter constant (see object 61A1h) and added to the previous calculated sum.

$$Value_N = Value_{N-1} + \frac{Input - Value_{N-1}}{Filterconstant}$$

- Calculation of repeating average**

At any reading of the current value from the A/D converter the value has to be added to the sum. At any N-th reading (depending on the value of object 61A1h) the sum will be divided by N. Following the value will be set to zero for the next calculation.

$$Value = \frac{\sum_{n=1}^N Input_n}{N}$$

#### OBJECT DESCRIPTION

|             |                |
|-------------|----------------|
| Index       | 61A0h          |
| Name        | AI Filter type |
| Object Code | ARRAY          |
| Data Type   | Unsigned8      |
| Category    | Optional       |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                  |
|----------------|------------------|
| Sub-Index      | 1h               |
| Description    | AI Filter type 1 |
| Entry Category | Mandatory        |
| Access         | rw               |
| PDO Mapping    | No               |
| Value Range    | Unsigned8        |
| Default Value  | 0                |

|                |                  |
|----------------|------------------|
| Sub-Index      | 2h               |
| Description    | AI Filter type 2 |
| Entry Category | Optional         |
| Access         | rw               |
| PDO Mapping    | No               |
| Value Range    | Unsigned8        |
| Default Value  | 0                |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AI Filter type 199 |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | Unsigned8          |
| Default Value  | 0                  |

**7.2.37 Object 61A1h: AI Filter constant**

Defines the number of steps used for the filter calculation (see Object 61A0h: AI Filter type).

## OBJECT DESCRIPTION

|             |                    |
|-------------|--------------------|
| Index       | 61A1h              |
| Name        | AI Filter constant |
| Object Code | ARRAY              |
| Data Type   | Unsigned8          |
| Category    | Optional           |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 1h                   |
| Description    | AI Filter constant 1 |
| Entry Category | Mandatory            |
| Access         | rw                   |
| PDO Mapping    | No                   |
| Value Range    | Unsigned8            |
| Default Value  | 0                    |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 2h                   |
| Description    | AI Filter constant 2 |
| Entry Category | Optional             |
| Access         | rw                   |
| PDO Mapping    | No                   |
| Value Range    | Unsigned8            |
| Default Value  | 0                    |

to

|                |                        |
|----------------|------------------------|
| Sub-Index      | C7h                    |
| Description    | AI Filter constant 199 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | Unsigned8              |
| Default Value  | 0                      |

**7.2.38 Object 6200h: DO Write state 8 output lines**

This object shall set a group of 8 output lines as a byte of information. A maximum of 199 x 8-bit output blocks is addressable.

## OBJECT DESCRIPTION

|             |                               |
|-------------|-------------------------------|
| Index       | 6200h                         |
| Name        | DO Write state 8 output lines |
| Object Code | ARRAY                         |
| Data Type   | Unsigned8                     |
| Category    | Conditional                   |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | 1h                              |
| Description    | DO Write state 8 output lines 1 |
| Entry Category | Mandatory                       |
| Access         | rw                              |
| PDO Mapping    | Possible                        |
| Value Range    | Unsigned8                       |
| Default Value  | 0                               |

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | 2h                              |
| Description    | DO Write state 8 output lines 2 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | Possible                        |
| Value Range    | Unsigned8                       |
| Default Value  | 0                               |

to

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | C7h                               |
| Description    | DO Write state 8 output lines 199 |
| Entry Category | Optional                          |
| Access         | rw                                |
| PDO Mapping    | Possible                          |
| Value Range    | Unsigned8                         |
| Default Value  | 0                                 |

**7.2.39 Object 6202h: DO Polarity 8 output lines**

This object shall define the polarity of a group of 8 output lines. Output polarity can be inverted individually.

0 – output not inverted                      1 – output inverted

If the object is not supported the device behaves accordingly to the default value.

## OBJECT DESCRIPTION

|             |                                   |
|-------------|-----------------------------------|
| Index       | 6202h                             |
| Name        | <b>DO Polarity 8 output lines</b> |
| Object Code | ARRAY                             |
| Data Type   | Unsigned8                         |
| Category    | Optional                          |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 1h                                  |
| Description    | <b>DO Polarity 8 output lines 1</b> |
| Entry Category | Mandatory                           |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | Unsigned8                           |
| Default Value  | 0                                   |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 2h                                  |
| Description    | <b>DO Polarity 8 output lines 2</b> |
| Entry Category | Optional                            |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | Unsigned8                           |
| Default Value  | 0                                   |

to

|                |                                       |
|----------------|---------------------------------------|
| Sub-Index      | C7h                                   |
| Description    | <b>DO Polarity 8 output lines 199</b> |
| Entry Category | Optional                              |
| Access         | rw                                    |
| PDO Mapping    | No                                    |
| Value Range    | Unsigned8                             |
| Default Value  | 0                                     |

**7.2.40 Object 6206h: DO Fault mode 8 output lines**

Defines the fault mode of a group of 8 output lines (1 bit per line).

0 – output value shall be kept if an fault occurs

1 – output value shall take the pre-defined condition specified in Object 6207h: DO Fault state 8 output lines

## OBJECT DESCRIPTION

|             |                              |
|-------------|------------------------------|
| Index       | 6206h                        |
| Name        | DO Fault mode 8 output lines |
| Object Code | ARRAY                        |
| Data Type   | Unsigned8                    |
| Category    | Optional                     |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                |
|----------------|--------------------------------|
| Sub-Index      | 1h                             |
| Description    | DO Fault mode 8 output lines 1 |
| Entry Category | Mandatory                      |
| Access         | rw                             |
| PDO Mapping    | No                             |
| Value Range    | Unsigned8                      |
| Default Value  | FFh                            |

|                |                                |
|----------------|--------------------------------|
| Sub-Index      | 2h                             |
| Description    | DO Fault mode 8 output lines 2 |
| Entry Category | Optional                       |
| Access         | rw                             |
| PDO Mapping    | No                             |
| Value Range    | Unsigned8                      |
| Default Value  | FFh                            |

to

|                |                                  |
|----------------|----------------------------------|
| Sub-Index      | C7h                              |
| Description    | DO Fault mode 8 output lines 199 |
| Entry Category | Optional                         |
| Access         | rw                               |
| PDO Mapping    | No                               |
| Value Range    | Unsigned8                        |
| Default Value  | FFh                              |

**7.2.41 Object 6207h: DO Fault state 8 output lines**

Defines the pre-defined state on detecting a fault condition. Defined for a group of 8 output lines. The corresponding bit must be set in the default output line mode (see Object 6206h: DO Fault mode 8 output lines).

- 0 – output is set to '0' in case of fault, if Object 6206h: DO Fault mode 8 output lines is enabled
- 1 – output is set to '1' in case of fault, if Object 6206h: DO Fault mode 8 output lines is enabled

## OBJECT DESCRIPTION

|             |                                      |
|-------------|--------------------------------------|
| Index       | 6207h                                |
| Name        | <b>DO Fault state 8 output lines</b> |
| Object Code | ARRAY                                |
| Data Type   | Unsigned8                            |
| Category    | Optional                             |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 1h                                     |
| Description    | <b>DO Fault state 8 output lines 1</b> |
| Entry Category | Mandatory                              |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Unsigned8                              |
| Default Value  | 0                                      |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 2h                                     |
| Description    | <b>DO Fault state 8 output lines 2</b> |
| Entry Category | Optional                               |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Unsigned8                              |
| Default Value  | 0                                      |

to

|                |                                          |
|----------------|------------------------------------------|
| Sub-Index      | C7h                                      |
| Description    | <b>DO Fault state 8 output lines 199</b> |
| Entry Category | Optional                                 |
| Access         | rw                                       |
| PDO Mapping    | No                                       |
| Value Range    | Unsigned8                                |
| Default Value  | 0                                        |

**7.2.42 Object 6210h: DO Link object 8 output lines**

This object describes the source for the DO Write state 8 output lines if it lies within another functional block of the same device. The object defined by index and sub-index is directly linked to the DO Write state 8 output lines.

## OBJECT DESCRIPTION

|             |                                      |
|-------------|--------------------------------------|
| Index       | 6210h                                |
| Name        | <b>DO Link object 8 output lines</b> |
| Object Code | RECORD                               |
| Data Type   | PDOmapping                           |
| Category    | Optional                             |



## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 1h                                     |
| Description    | <b>DO Link object 8 output lines 1</b> |
| Entry Category | Mandatory                              |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Unsigned32                             |
| Default Value  | No                                     |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 2h                                     |
| Description    | <b>DO Link object 8 output lines 2</b> |
| Entry Category | Optional                               |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Unsigned32                             |
| Default Value  | No                                     |

to

|                |                                          |
|----------------|------------------------------------------|
| Sub-Index      | C7h                                      |
| Description    | <b>DO Link object 8 output lines 199</b> |
| Entry Category | Optional                                 |
| Access         | rw                                       |
| PDO Mapping    | No                                       |
| Value Range    | Unsigned32                               |
| Default Value  | No                                       |

**7.2.43 Object 6220h: DO Write state 1 output lines**

This object shall set single output lines information. A maximum of 199 outputs is addressable at this index.

## OBJECT DESCRIPTION

|             |                               |
|-------------|-------------------------------|
| Index       | 6220h                         |
| Name        | DO Write state 1 output lines |
| Object Code | ARRAY                         |
| Data Type   | Boolean                       |
| Category    | Optional                      |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | 1h                              |
| Description    | DO Write state 1 output lines 1 |
| Entry Category | Mandatory                       |
| Access         | rw                              |
| PDO Mapping    | Possible                        |
| Value Range    | Boolean                         |
| Default Value  | 0                               |

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | 2h                              |
| Description    | DO Write state 1 output lines 2 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | Possible                        |
| Value Range    | Boolean                         |
| Default Value  | 0                               |

to

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | C7h                               |
| Description    | DO Write state 1 output lines 199 |
| Entry Category | Optional                          |
| Access         | rw                                |
| PDO Mapping    | Possible                          |
| Value Range    | Boolean                           |
| Default Value  | 0                                 |

**7.2.44 Object 6240h: DO Polarity 1 output lines**

This object shall set the polarity of single output lines. A maximum of 199 outputs is addressable at this index.

0 – output inverted

1 – output not inverted

If this object is not supported the device shall behave accordingly to the default value.

## OBJECT DESCRIPTION

|             |                                   |
|-------------|-----------------------------------|
| Index       | 6240h                             |
| Name        | <b>DO Polarity 1 output lines</b> |
| Object Code | ARRAY                             |
| Data Type   | Boolean                           |
| Category    | Optional                          |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 1h                                  |
| Description    | <b>DO Polarity 1 output lines 1</b> |
| Entry Category | Mandatory                           |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | Boolean                             |
| Default Value  | 0                                   |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 2h                                  |
| Description    | <b>DO Polarity 1 output lines 2</b> |
| Entry Category | Optional                            |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | Boolean                             |
| Default Value  | 0                                   |

to

|                |                                       |
|----------------|---------------------------------------|
| Sub-Index      | C7h                                   |
| Description    | <b>DO Polarity 1 output lines 199</b> |
| Entry Category | Optional                              |
| Access         | rw                                    |
| PDO Mapping    | No                                    |
| Value Range    | Boolean                               |
| Default Value  | 0                                     |

**7.2.45 Object 6250h: DO Fault mode 1 output lines**

Sets the fault mode of a single output line. This defines the default output mode on detecting a fault condition.

0 – output value shall be kept if a fault occurs

1 – output value shall take the pre-defined condition as specified in Object 6260h: DO Fault state 1 output lines

## OBJECT DESCRIPTION

|             |                                     |
|-------------|-------------------------------------|
| Index       | 6250h                               |
| Name        | <b>DO Fault mode 1 output lines</b> |
| Object Code | ARRAY                               |
| Data Type   | Boolean                             |
| Category    | Optional                            |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                       |
|----------------|---------------------------------------|
| Sub-Index      | 1h                                    |
| Description    | <b>DO Fault mode 1 output lines 1</b> |
| Entry Category | Mandatory                             |
| Access         | rw                                    |
| PDO Mapping    | No                                    |
| Value Range    | Boolean                               |
| Default Value  | True                                  |

|                |                                       |
|----------------|---------------------------------------|
| Sub-Index      | 2h                                    |
| Description    | <b>DO Fault mode 1 output lines 2</b> |
| Entry Category | Optional                              |
| Access         | rw                                    |
| PDO Mapping    | No                                    |
| Value Range    | Boolean                               |
| Default Value  | True                                  |

to

|                |                                         |
|----------------|-----------------------------------------|
| Sub-Index      | C7h                                     |
| Description    | <b>DO Fault mode 1 output lines 199</b> |
| Entry Category | Optional                                |
| Access         | rw                                      |
| PDO Mapping    | No                                      |
| Value Range    | Boolean                                 |
| Default Value  | True                                    |

**7.2.46 Object 6260h: DO Fault state 1 output lines**

Output line fault state. This defines the pre-defined output state on detecting a fault condition. The corresponding bit must be set in the default output line mode (see Object 6250h: DO Fault mode 1 output lines).

- 0 – output shall be set to '0' in case of fault, if the corresponding Object 6250h: DO Fault mode 1 output lines is enabled
- 1 – output shall be set to '1' in case of fault, if the corresponding Object 6250h: DO Fault mode 1 output lines is enabled

## OBJECT DESCRIPTION

|             |                                      |
|-------------|--------------------------------------|
| Index       | 6260h                                |
| Name        | <b>DO Fault state 1 output lines</b> |
| Object Code | ARRAY                                |
| Data Type   | Boolean                              |
| Category    | Optional                             |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 1h                                     |
| Description    | <b>DO Fault state 1 output lines 1</b> |
| Entry Category | Mandatory                              |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Boolean                                |
| Default Value  | 0                                      |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 2h                                     |
| Description    | <b>DO Fault state 1 output lines 2</b> |
| Entry Category | Optional                               |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Boolean                                |
| Default Value  | 0                                      |

to

|                |                                          |
|----------------|------------------------------------------|
| Sub-Index      | C7h                                      |
| Description    | <b>DO Fault state 1 output lines 199</b> |
| Entry Category | Optional                                 |
| Access         | rw                                       |
| PDO Mapping    | No                                       |
| Value Range    | Boolean                                  |
| Default Value  | 0                                        |

**7.2.47 Object 6280h: DO Link object 1 output lines**

This object describes the source for the DO Write state 1 output lines if it lies within another functional block of the same device. The object defined by index and Sub-Index is directly linked to the DO Write state 1 output lines.

## OBJECT DESCRIPTION

|             |                                      |
|-------------|--------------------------------------|
| Index       | 6280h                                |
| Name        | <b>DO Link object 1 output lines</b> |
| Object Code | RECORD                               |
| Data Type   | PDOmapping                           |
| Category    | Optional                             |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – C7h           |
| Default Value  | No                |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 1h                                     |
| Description    | <b>DO Link object 1 output lines 1</b> |
| Entry Category | Mandatory                              |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Unsigned32                             |
| Default Value  | No                                     |

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | 2h                                     |
| Description    | <b>DO Link object 1 output lines 2</b> |
| Entry Category | Optional                               |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Unsigned32                             |
| Default Value  | No                                     |

to

|                |                                          |
|----------------|------------------------------------------|
| Sub-Index      | C7h                                      |
| Description    | <b>DO Link object 1 output lines 199</b> |
| Entry Category | Optional                                 |
| Access         | rw                                       |
| PDO Mapping    | No                                       |
| Value Range    | Unsigned32                               |
| Default Value  | No                                       |

**7.2.48 Object x300h: AO Output PV**

This object represents the process value, which is the input to the analogue output function block. It is scaled in the physical unit of process values (e.g. degrees centigrade, kg, kN, mm etc.).

## OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x300h                                  |
| Name        | AO Output PV                           |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Conditional                            |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AO Output PV 1     |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | 0                  |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AO Output PV 2     |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | 0                  |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AO Output PV 199   |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | 0                  |

**7.2.49 Object 6301h: AO Physical unit PV**

This object describes the physical unit for the process values within the analogue output function block (see Object 6131h: AI Physical unit PV).

## OBJECT DESCRIPTION

|             |                     |
|-------------|---------------------|
| Index       | 6301h               |
| Name        | AO Physical unit PV |
| Object Code | ARRAY               |
| Data Type   | Unsigned32          |
| Category    | Optional            |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 1h                    |
| Description    | AO Physical unit PV 1 |
| Entry Category | Mandatory             |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned16            |
| Default Value  | No                    |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 2h                    |
| Description    | AO Physical unit PV 2 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned16            |
| Default Value  | No                    |

to

|                |                         |
|----------------|-------------------------|
| Sub-Index      | C7h                     |
| Description    | AO Physical unit PV 199 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | Unsigned16              |
| Default Value  | No                      |

**7.2.50 Object 6302h: AO Decimal digits PV**

This object describes the number of decimal digits following the decimal point for interpretation of data types Integer8, Integer16 and Integer32.

Example: A field value of 1.230 (Float) will be coded as 123 in Integer16 format if number of decimal digits is set to 2.

## OBJECT DESCRIPTION

|             |                      |
|-------------|----------------------|
| Index       | 6302h                |
| Name        | AO Decimal digits PV |
| Object Code | ARRAY                |
| Data Type   | Unsigned8            |
| Category    | Optional             |



## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | AO Decimal digits PV 1 |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 9h                 |
| Default Value  | No                     |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | AO Decimal digits PV 2 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 9h                 |
| Default Value  | No                     |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | AO Decimal digits PV 199 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 – 9h                   |
| Default Value  | No                       |

**7.2.51 Object 6303h: AO Link output PV**

This object describes the source for the AO Output PV if it lies within another functional block of the same device. The object defined by index and sub-index is directly linked to the AO Output PV.

## OBJECT DESCRIPTION

|             |                   |
|-------------|-------------------|
| Index       | 6303h             |
| Name        | AO Link output PV |
| Object Code | RECORD            |
| Data Type   | PDOmapping        |
| Category    | Optional          |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - 199           |
| Default Value  | No                |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 1h                  |
| Description    | AO Link output PV 1 |
| Entry Category | Mandatory           |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | Unsigned32          |
| Default Value  | No                  |

|                |                     |
|----------------|---------------------|
| Sub-Index      | 2h                  |
| Description    | AO Link output PV 2 |
| Entry Category | Optional            |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | Unsigned32          |
| Default Value  | No                  |

to

|                |                       |
|----------------|-----------------------|
| Sub-Index      | C7h                   |
| Description    | AO Link output PV 199 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned32            |
| Default Value  | No                    |

**7.2.52 Object 6310h: AO Output type**

This object specifies the type of the analogue output.

| Value    | Meaning   |
|----------|-----------|
| 0 .. 9   | reserved  |
| 10       | Voltage   |
| 11       | +/-10V    |
| 12 .. 19 | reserved  |
| 20       | Current   |
| 21       | 4..20mA   |
| 22       | +/-20mA   |
| 23 .. 29 | reserved  |
| 30       | Frequency |
| 31 .. 39 | reserved  |

| Value         | Meaning               |
|---------------|-----------------------|
| 40            | PWM                   |
| 41 ... 999    | reserved              |
| 1000 .. 65535 | manufacturer specific |

## OBJECT DESCRIPTION

|             |                       |
|-------------|-----------------------|
| Index       | 6310h                 |
| Name        | <b>AO Output type</b> |
| Object Code | ARRAY                 |
| Data Type   | Unsigned16            |
| Category    | Optional              |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | <b>AO Output type 1</b> |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | Unsigned16              |
| Default Value  | No                      |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | <b>AO Output type 2</b> |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | Unsigned16              |
| Default Value  | No                      |

to

|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | <b>AO Output type 199</b> |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | Unsigned16                |
| Default Value  | No                        |

**7.2.53 Object x320h: AO Output scaling 1 PV**

This object defines the process value of the first calibration point for the analogue output channel. It is scaled in physical unit of process value.

## OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x320h                                  |
| Name        | AO Output scaling 1 PV                 |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | AO Output scaling 1 PV 1 |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | AO Output scaling 1 PV 2 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | AO Output scaling 1 PV 199 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

**7.2.54 Object x321h: AO Output scaling 1 FV**

This object defines the field value of the first calibration point for the analogue output channel. It is scaled in physical unit of field value.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x321h                                  |
| Name        | AO Output scaling 1 FV                 |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | AO Output scaling 1 FV 1 |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | AO Output scaling 1 FV 2 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | AO Output scaling 1 FV 199 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

**7.2.55 Object x322h: AO Output scaling 2 PV**

This object defines the process value of the second calibration point for the analogue output channel. It is scaled in physical unit of process value.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x322h                                  |
| Name        | AO Output scaling 2 PV                 |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | AO Output scaling 2 PV 1 |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | AO Output scaling 2 PV 2 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | AO Output scaling 2 PV 199 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

**7.2.56 Object x323h: AO Output scaling 2 FV**

This object defines the field value of the second calibration point for the analogue output channel. It is scaled in physical unit of field value.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x323h                                  |
| Name        | AO Output scaling 2 FV                 |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | AO Output scaling 2 FV 1 |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | AO Output scaling 2 FV 2 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | AO Output scaling 2 FV 199 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

**7.2.57 Object x330h: AO Output FV**

This object represents result of the output scaling for the analogue output channel and gives the output quantity scaled in the physical unit of field values (e.g. Volt, mA etc.).

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x330h                                  |
| Name        | AO Output FV                           |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AO Output FV 1     |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AO Output FV 2     |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AO Output FV 199   |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |



**7.2.58 Object 6331h: AO Physical unit FV**

This object describes the physical unit for the field values within the analogue output function block (see Object 6131h: AI Physical unit PV).

**OBJECT DESCRIPTION**

|             |                     |
|-------------|---------------------|
| Index       | 6331h               |
| Name        | AO Physical unit FV |
| Object Code | ARRAY               |
| Data Type   | Unsigned32          |
| Category    | Optional            |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 1h                    |
| Description    | AO Physical unit FV 1 |
| Entry Category | Mandatory             |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned32            |
| Default Value  | No                    |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 2h                    |
| Description    | AO Physical unit FV 2 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned32            |
| Default Value  | No                    |

to

|                |                         |
|----------------|-------------------------|
| Sub-Index      | C7h                     |
| Description    | AO Physical unit FV 199 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | Unsigned32              |
| Default Value  | No                      |

**7.2.59 Object 6332h: AO Decimal digits FV**

This object describes the number of decimal digits following the decimal point for interpretation of data types Integer8, Integer16 and Integer32.

Example: A field value of 1.230 (Float) will be coded as 123 in Integer16 format if number of decimal digits is set to 2.

**OBJECT DESCRIPTION**

|             |                      |
|-------------|----------------------|
| Index       | 6332h                |
| Name        | AO Decimal digits FV |
| Object Code | ARRAY                |
| Data Type   | Unsigned8            |
| Category    | Optional             |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | N                 |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | AO Decimal digits FV 1 |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 9h                 |
| Default Value  | No                     |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | AO Decimal digits FV 2 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 9h                 |
| Default Value  | No                     |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | AO Decimal digits FV 199 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 – 9h                   |
| Default Value  | No                       |

**7.2.60 Object 6340h: AO Fault mode**

This object defines, whether an output is set to a pre-defined error value (see Object x341h: AO Fault FV) in case of an internal device failure.

- 0 – actual value rest
- 1 – reverts to error value (see Object x341h: AO Fault FV)
- other – reserved

**OBJECT DESCRIPTION**

|             |                      |
|-------------|----------------------|
| Index       | 6340h                |
| Name        | <b>AO Fault mode</b> |
| Object Code | ARRAY                |
| Data Type   | Unsigned8            |
| Category    | Optional             |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | <b>AO Fault mode 1</b> |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 1h                 |
| Default Value  | 1                      |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | <b>AO Fault mode 2</b> |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 – 1h                 |
| Default Value  | 1                      |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | <b>AO Fault mode 199</b> |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 – 1h                   |
| Default Value  | 1                        |

**7.2.61 Object x341h: AO Fault FV**

Field value to be taken as output if there is a fault condition and AO Fault Mode is 1.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x341h                                  |
| Name        | AO Fault FV                            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AO Fault FV 1      |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AO Fault FV 2      |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AO Fault FV 199    |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

**7.2.62 Object x400h: CO Effective Current Value Xeff**

This object defines the effective current value used by the control algorithm. The physical unit and the number of decimal digits of this object are defined by objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x400h                                  |
| Name        | CO Effective Current Value Xeff        |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Mandatory                              |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                              |
|----------------|------------------------------|
| Sub-Index      | 1h                           |
| Description    | CO Effective Current Value 1 |
| Entry Category | Mandatory                    |
| Access         | rw                           |
| PDO Mapping    | Possible                     |
| Value Range    | data type specific           |
| Default Value  | No                           |

|                |                              |
|----------------|------------------------------|
| Sub-Index      | 2h                           |
| Description    | CO Effective Current Value 2 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | Possible                     |
| Value Range    | data type specific           |
| Default Value  | No                           |

to

|                |                                |
|----------------|--------------------------------|
| Sub-Index      | C7h                            |
| Description    | CO Effective Current Value 199 |
| Entry Category | Optional                       |
| Access         | rw                             |
| PDO Mapping    | Possible                       |
| Value Range    | data type specific             |
| Default Value  | No                             |

### 7.2.63 Object x401h: CO Effective Set Point Weff

This object holds the set point currently used by the controller algorithm. In most cases it will be a copy of the objects x402h or x403h (it may be generated by set point ramp also). The physical unit and the number of decimal digits of this object are defined by objects 6406h and 6407h.

#### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x401h                                  |
| Name        | CO Effective Set Point Weff            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Mandatory                              |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | CO Effective Set Point 1 |
| Entry Category | Mandatory                |
| Access         | ro                       |
| PDO Mapping    | Possible                 |
| Value Range    | data type specific       |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | CO Effective Set Point 2 |
| Entry Category | Optional                 |
| Access         | ro                       |
| PDO Mapping    | Possible                 |
| Value Range    | data type specific       |
| Default Value  | No                       |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | Effective Set Point Weff 199 |
| Entry Category | Optional                     |
| Access         | ro                           |
| PDO Mapping    | Possible                     |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.64 Object x402h: CO Set point W**

The first set point is variable within the limits of W0 and W100 ( $W0 \leq W \leq W100$ ). The physical unit and the number of decimal digits of this object are defined by objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x402h                                  |
| Name        | CO Set point W                         |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Mandatory                              |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | CO Set point W 1   |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | CO Set point W 2   |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | CO Set point W 199 |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

**7.2.65 Object x403h: CO 2nd Set point W2**

The second set point is variable within the limits of W0 and W100 ( $W0 \leq W \leq W100$ ). The physical unit and the number of decimal digits of this object are defined by objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x403h                                  |
| Name        | CO 2nd Set point W2                    |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Mandatory                              |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | CO 2nd Set point 1 |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | CO 2nd Set point 2 |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | data type specific |
| Default Value  | No                 |

to

|                |                      |
|----------------|----------------------|
| Sub-Index      | C7h                  |
| Description    | CO 2nd Set point 199 |
| Entry Category | Optional             |
| Access         | rw                   |
| PDO Mapping    | Possible             |
| Value Range    | data type specific   |
| Default Value  | No                   |



**7.2.66 Object x404h: CO Lower Set Point Limit W0**

The object describes the lower limit of the set point. The physical unit and the number of decimal digits of this object are defined by objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x404h                                  |
| Name        | CO Lower Set Point Limit W0            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | CO Lower Set Point Limit 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | CO Lower Set Point Limit 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | CO Lower Set Point Limit 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.67 Object x405h: CO Upper Set Point Limit W100**

The object describes the upper limit of the set point. The physical unit and the number of decimal digits of this object are defined by objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x405h                                  |
| Name        | CO Upper Set Point Limit W100          |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | CO Upper Set Point Limit 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | CO Upper Set Point Limit 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | CO Upper Set Point Limit 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.68 Object 6406h: CO Physical Unit Current Value / Set Point**

The object describes the physical unit of the objects beginning from x400h to x405h (see Object 6131h: AI Physical unit PV).

**OBJECT DESCRIPTION**

|             |                     |
|-------------|---------------------|
| Index       | 6406h               |
| Name        | CO Physical Unit XW |
| Object Code | ARRAY               |
| Data Type   | Unsigned32          |
| Category    | Mandatory           |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 1h                    |
| Description    | CO Physical Unit XW 1 |
| Entry Category | Mandatory             |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned32            |
| Default Value  | No                    |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 2h                    |
| Description    | CO Physical Unit XW 2 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | Unsigned32            |
| Default Value  | No                    |

to

|                |                         |
|----------------|-------------------------|
| Sub-Index      | C7h                     |
| Description    | CO Physical Unit XW 199 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | Unsigned32              |
| Default Value  | No                      |

**7.2.69 Object 6407h: CO Decimal Digits Current Value / Set Point**

The object describes the number of decimal digits of the objects beginning from x400h to x405h.

**OBJECT DESCRIPTION**

|             |                      |
|-------------|----------------------|
| Index       | 6407h                |
| Name        | CO Decimal Digits XW |
| Object Code | ARRAY                |
| Data Type   | Unsigned8            |
| Category    | Mandatory            |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | CO Decimal Digits XW 1 |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 - 15                 |
| Default Value  | No                     |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | CO Decimal Digits XW 2 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 - 15                 |
| Default Value  | No                     |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | CO Decimal Digits XW 199 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 - 15                   |
| Default Value  | No                       |

**7.2.70 Object 6410h: CO Effective Controller Output Y**

This object holds effective controller output calculated by the PID algorithm. The object is read only. The value is given in percent with one decimal digit after the comma. The value will always be within the range of Ymin and Ymax ( $Y_{min} \leq Y \leq Y_{max}$ ).

| hexadecimal value | decimal value | percent value |
|-------------------|---------------|---------------|
| FC18h             | -1000         | -100.0 %      |
| .....             | .....         | .....         |
| FE0Ch             | -500          | -50.0 %       |
| .....             | .....         | .....         |
| 0000h             | 0             | 0.0 %         |
| .....             | .....         | .....         |
| 01F4h             | 500           | 50.0 %        |
| .....             | .....         | .....         |
| 03E8h             | 1000          | 100.0 %       |

**OBJECT DESCRIPTION**

|             |                                  |
|-------------|----------------------------------|
| Index       | 6410h                            |
| Name        | CO Effective Controller Output Y |
| Object Code | ARRAY                            |
| Data Type   | Unsigned16                       |
| Category    | Mandatory                        |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                  |
|----------------|----------------------------------|
| Sub-Index      | 1h                               |
| Description    | CO Effective Controller Output 1 |
| Entry Category | Mandatory                        |
| Access         | ro                               |
| PDO Mapping    | Possible                         |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh     |
| Default Value  | No                               |

|                |                                  |
|----------------|----------------------------------|
| Sub-Index      | 2h                               |
| Description    | CO Effective Controller Output 2 |
| Entry Category | Optional                         |
| Access         | ro                               |
| PDO Mapping    | Possible                         |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh     |
| Default Value  | No                               |

to

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | C7h                                |
| Description    | CO Effective Controller Output 199 |
| Entry Category | Optional                           |
| Access         | ro                                 |
| PDO Mapping    | Possible                           |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh       |
| Default Value  | No                                 |

### 7.2.71 Object 6411h: CO Differential Controller Output Ydiff

This object holds differential controller output calculated by the PID algorithm. The object is read only. The value is given in percent with one decimal digit after the comma (see Object 6410h: CO Effective Controller Output Y). The value will always be within the range of Ymin and Ymax ( $Y_{diff} < (Y_{max} - Y_{min})$ ).

#### OBJECT DESCRIPTION

|             |                                      |
|-------------|--------------------------------------|
| Index       | 6411j                                |
| Name        | Differential Controller Output Ydiff |
| Object Code | ARRAY                                |
| Data Type   | Unsigned16                           |
| Category    | Optional                             |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 1h                                  |
| Description    | CO Differential Controller Output 1 |
| Entry Category | Mandatory                           |
| Access         | ro                                  |
| PDO Mapping    | Possible                            |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh        |
| Default Value  | No                                  |

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | 2h                                  |
| Description    | CO Differential Controller Output 2 |
| Entry Category | Optional                            |
| Access         | ro                                  |
| PDO Mapping    | Possible                            |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh        |
| Default Value  | No                                  |

to

|                |                                       |
|----------------|---------------------------------------|
| Sub-Index      | C7h                                   |
| Description    | CO Differential Controller Output 199 |
| Entry Category | Optional                              |
| Access         | ro                                    |
| PDO Mapping    | Possible                              |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh          |
| Default Value  | No                                    |

### 7.2.72 Object 6412h: CO Manual Controller Output

The value is used as output in the manual controller mode. The value is given in percent with one decimal digit after the comma (see Object 6410h: CO Effective Controller Output Y).

#### OBJECT DESCRIPTION

|             |                          |
|-------------|--------------------------|
| Index       | 6412h                    |
| Name        | Manual Controller Output |
| Object Code | ARRAY                    |
| Data Type   | Unsigned16               |
| Category    | Optional                 |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                              |
|----------------|------------------------------|
| Sub-Index      | 1h                           |
| Description    | Manual Controller Output 1   |
| Entry Category | Mandatory                    |
| Access         | rw                           |
| PDO Mapping    | Possible                     |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh |
| Default Value  | 0                            |

|                |                              |
|----------------|------------------------------|
| Sub-Index      | 2h                           |
| Description    | Manual Controller Output 2   |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | Possible                     |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh |
| Default Value  | 0                            |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | Manual Controller Output 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | Possible                     |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh |
| Default Value  | 0                            |

### 7.2.73 Object 6413h: CO Lower Controller Output Limit Ymin

The value describes the lower limit for the controller output. The value is given in percent with one decimal digit after the comma (see Object 6410h: CO Effective Controller Output Y).

#### OBJECT DESCRIPTION

|             |                                    |
|-------------|------------------------------------|
| Index       | 6413h                              |
| Name        | Lower Controller Output Limit Ymin |
| Object Code | ARRAY                              |
| Data Type   | Unsigned16                         |
| Category    | Optional                           |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 1h                                 |
| Description    | CO Lower Controller Output Limit 1 |
| Entry Category | Mandatory                          |
| Access         | rw                                 |
| PDO Mapping    | No                                 |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh       |
| Default Value  | No                                 |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 2h                                 |
| Description    | CO Lower Controller Output Limit 2 |
| Entry Category | Optional                           |
| Access         | rw                                 |
| PDO Mapping    | No                                 |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh       |
| Default Value  | No                                 |

to



|                |                                      |
|----------------|--------------------------------------|
| Sub-Index      | C7h                                  |
| Description    | CO Lower Controller Output Limit 199 |
| Entry Category | Optional                             |
| Access         | rw                                   |
| PDO Mapping    | No                                   |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh         |
| Default Value  | No                                   |

#### 7.2.74 Object 6414h: CO Upper Controller Output Limit Ymax

The value describes the lower limit for the controller output. The value is given in percent with one decimal digit after the comma (see Object 6410h: CO Effective Controller Output Y).

##### OBJECT DESCRIPTION

|             |                                    |
|-------------|------------------------------------|
| Index       | 6414h                              |
| Name        | Upper Controller Output Limit Ymax |
| Object Code | ARRAY                              |
| Data Type   | Unsigned16                         |
| Category    | Optional                           |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 1h                                 |
| Description    | CO Upper Controller Output Limit 1 |
| Entry Category | Mandatory                          |
| Access         | rw                                 |
| PDO Mapping    | No                                 |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh       |
| Default Value  | No                                 |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 2h                                 |
| Description    | CO Upper Controller Output Limit 2 |
| Entry Category | Optional                           |
| Access         | rw                                 |
| PDO Mapping    | No                                 |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh       |
| Default Value  | No                                 |

to

|                |                                      |
|----------------|--------------------------------------|
| Sub-Index      | C7h                                  |
| Description    | CO Upper Controller Output Limit 199 |
| Entry Category | Optional                             |
| Access         | rw                                   |
| PDO Mapping    | No                                   |
| Value Range    | 0000h – 03E8h, FC18h - FFFFh         |
| Default Value  | No                                   |

### 7.2.75 Object 6415h: CO Physical Unit Controller Output

The object describes the physical unit of the objects beginning from 6410h to 6414h (see Object 6131h: AI Physical unit PV).

#### OBJECT DESCRIPTION

|             |                                    |
|-------------|------------------------------------|
| Index       | 6415h                              |
| Name        | CO Physical Unit Controller Output |
| Object Code | ARRAY                              |
| Data Type   | Unsigned32                         |
| Category    | Mandatory                          |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                      |
|----------------|--------------------------------------|
| Sub-Index      | 1h                                   |
| Description    | CO Physical Unit Controller Output 1 |
| Entry Category | Mandatory                            |
| Access         | rw                                   |
| PDO Mapping    | No                                   |
| Value Range    | Unsigned32                           |
| Default Value  | No                                   |

|                |                                      |
|----------------|--------------------------------------|
| Sub-Index      | 2h                                   |
| Description    | CO Physical Unit Controller Output 2 |
| Entry Category | Optional                             |
| Access         | rw                                   |
| PDO Mapping    | No                                   |
| Value Range    | Unsigned32                           |
| Default Value  | No                                   |

to

|                |                                        |
|----------------|----------------------------------------|
| Sub-Index      | C7h                                    |
| Description    | CO Physical Unit Controller Output 199 |
| Entry Category | Optional                               |
| Access         | rw                                     |
| PDO Mapping    | No                                     |
| Value Range    | Unsigned32                             |
| Default Value  | No                                     |

#### 7.2.76 Object 6420h: CO Set Point switch W/W2

Set Point Switch W/W2 for switching between first and second set point.

FALSE – selects the first set point

TRUE – selects the second set point

##### OBJECT DESCRIPTION

|             |                       |
|-------------|-----------------------|
| Index       | 6420h                 |
| Name        | Set Point Switch W/W2 |
| Object Code | ARRAY                 |
| Data Type   | Boolean               |
| Category    | Mandatory             |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | Set Point Switch 1 |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | Boolean            |
| Default Value  | FALSE              |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | Set Point Switch 2 |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | Possible           |
| Value Range    | Boolean            |
| Default Value  | FALSE              |

to

|                |                      |
|----------------|----------------------|
| Sub-Index      | C7h                  |
| Description    | Set Point Switch 199 |
| Entry Category | Optional             |
| Access         | rw                   |
| PDO Mapping    | Possible             |
| Value Range    | Boolean              |
| Default Value  | FALSE                |

### 7.2.77 Object 6421h: CO Automatic/Manual Mode switch A/M

Switches between automatic and manual mode.

FALSE – selects the automatic mode

TRUE – selects the manual mode

#### OBJECT DESCRIPTION

|             |                           |
|-------------|---------------------------|
| Index       | 6421h                     |
| Name        | Automatic Manual Mode A/M |
| Object Code | ARRAY                     |
| Data Type   | Boolean                   |
| Category    | Mandatory                 |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | Automatic Manual Mode 1 |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | Possible                |
| Value Range    | Boolean                 |
| Default Value  | FALSE                   |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | Automatic Manual Mode 2 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | Possible                |
| Value Range    | Boolean                 |
| Default Value  | FALSE                   |

to

|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | Automatic Manual Mode 199 |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | Possible                  |
| Value Range    | Boolean                   |
| Default Value  | FALSE                     |

### 7.2.78 Object 6422h: CO Controller On/Off

Switches the controller on or off.

TRUE – switches the controller on

FALSE – switches the controller off

#### OBJECT DESCRIPTION

|             |                    |
|-------------|--------------------|
| Index       | 6412h              |
| Name        | Controller On/ Off |
| Object Code | ARRAY              |
| Data Type   | Boolean            |
| Category    | Mandatory          |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 1h                   |
| Description    | Controller On /Off 1 |
| Entry Category | Mandatory            |
| Access         | rw                   |
| PDO Mapping    | Possible             |
| Value Range    | Boolean              |
| Default Value  | FALSE                |

|                |                      |
|----------------|----------------------|
| Sub-Index      | 2h                   |
| Description    | Controller On/ Off 2 |
| Entry Category | Optional             |
| Access         | rw                   |
| PDO Mapping    | Possible             |
| Value Range    | Boolean              |
| Default Value  | FALSE                |

to

|                |                        |
|----------------|------------------------|
| Sub-Index      | C7h                    |
| Description    | Controller On /Off 199 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | Possible               |
| Value Range    | Boolean                |
| Default Value  | FALSE                  |

### 7.2.79 Object 6423h: CO Controller mode

Selects the mode of the controller. The meaning is shown below.

| control byte | Mode                                  |
|--------------|---------------------------------------|
| 00h          | 2-point controller                    |
| 01h          | 3-point controller                    |
| 02h          | 3-point stepping controller           |
| 03h          | continuous controller                 |
| 04h          | reserved for future use               |
| ...          | ...                                   |
| 7Fh          | reserved for future use               |
| 80h          | manufacturer specific controller type |
| ...          | ...                                   |
| FFh          | manufacturer specific controller type |

#### OBJECT DESCRIPTION

|             |                 |
|-------------|-----------------|
| Index       | 6423h           |
| Name        | Controller Mode |
| Object Code | ARRAY           |
| Data Type   | Unsigned8       |
| Category    | Mandatory       |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                   |
|----------------|-------------------|
| Sub-Index      | 1h                |
| Description    | Controller Mode 1 |
| Entry Category | Mandatory         |
| Access         | rw                |
| PDO Mapping    | Possible          |
| Value Range    | Unsigned8         |
| Default Value  | No                |

|                |                   |
|----------------|-------------------|
| Sub-Index      | 2h                |
| Description    | Controller Mode 2 |
| Entry Category | Optional          |
| Access         | rw                |
| PDO Mapping    | Possible          |
| Value Range    | Unsigned8         |
| Default Value  | No                |

to

|                |                     |
|----------------|---------------------|
| Sub-Index      | C7h                 |
| Description    | Controller Mode 199 |
| Entry Category | Optional            |
| Access         | rw                  |
| PDO Mapping    | Possible            |
| Value Range    | Unsigned8           |
| Default Value  | No                  |

#### 7.2.80 Object 6424h: CO Self Optimization on/off

This object switches the self-optimisation mode on or off.

FALSE – switches the self-optimisation off

TRUE – switches the self-optimisation on

##### OBJECT DESCRIPTION

|             |                             |
|-------------|-----------------------------|
| Index       | 6424h                       |
| Name        | CO Self Optimisation On/Off |
| Object Code | ARRAY                       |
| Data Type   | Boolean                     |
| Category    | Optional                    |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | CO Self Optimization 1 |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | Possible               |
| Value Range    | Boolean                |
| Default Value  | FALSE                  |

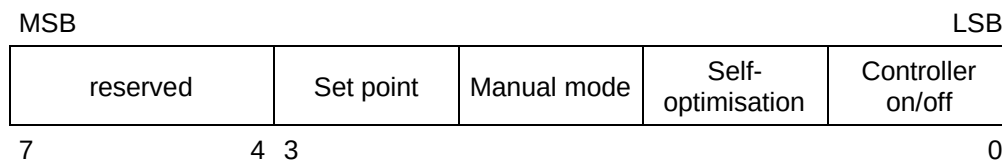
|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | CO Self Optimization 2 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | Possible               |
| Value Range    | Boolean                |
| Default Value  | FALSE                  |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | CO Self Optimization 199 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | Possible                 |
| Value Range    | Boolean                  |
| Default Value  | FALSE                    |

### 7.2.81 Object 6425h: CO Control Byte

The control byte maps the objects for set-point switch, controller on/off etc. in one single object. The main purpose for this mapping is to have a single object for the PDO mapping.



| Value | Meaning          |
|-------|------------------|
| 0     | disable function |
| 1     | enable function  |

#### OBJECT DESCRIPTION

|             |                 |
|-------------|-----------------|
| Index       | 6425h           |
| Name        | CO Control Byte |
| Object Code | ARRAY           |
| Data Type   | Unsigned8       |
| Category    | Mandatory       |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |



|                |                   |
|----------------|-------------------|
| Sub-Index      | 1h                |
| Description    | CO Control Byte 1 |
| Entry Category | Mandatory         |
| Access         | rw                |
| PDO Mapping    | Possible          |
| Value Range    | Unsigned8         |
| Default Value  | 0                 |

|                |                   |
|----------------|-------------------|
| Sub-Index      | 2h                |
| Description    | CO Control Byte 2 |
| Entry Category | Optional          |
| Access         | rw                |
| PDO Mapping    | Possible          |
| Value Range    | Unsigned8         |
| Default Value  | 0                 |

to

|                |                     |
|----------------|---------------------|
| Sub-Index      | C7h                 |
| Description    | CO Control Byte 199 |
| Entry Category | Optional            |
| Access         | rw                  |
| PDO Mapping    | Possible            |
| Value Range    | Unsigned8           |
| Default Value  | 0                   |

#### 7.2.82 Object 6426h: CO Control byte enable

This bit mask enables the single bits within the CO Control byte.

##### OBJECT DESCRIPTION

|             |                        |
|-------------|------------------------|
| Index       | 6426h                  |
| Name        | CO Control Byte Enable |
| Object Code | ARRAY                  |
| Data Type   | Unsigned8              |
| Category    | Optional               |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | CO Control Byte Enable 1 |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | Unsigned8                |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | CO Control Byte Enable 2 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | Unsigned8                |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | CO Control Byte Enable 199 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | Unsigned8                  |
| Default Value  | No                         |

### 7.2.83 Object 6427h: CO Status Word

The status word holds the status of the controller. The main purpose for this object is to have a single object for the PDO mapping.

The single bits of the CO status information have the following meaning:

| MSB      |  |               |           |                |          |   | LSB                |           |             |                   |                   |   |
|----------|--|---------------|-----------|----------------|----------|---|--------------------|-----------|-------------|-------------------|-------------------|---|
| reserved |  | Net over-load | Over-load | Data not valid | reserved |   | Optimisation error | Set point | Manual mode | Self-optimisation | Controller on/off |   |
| 15       |  | 11            | 10        | 9              | 8        | 7 | 5                  | 4         | 3           | 2                 | 1                 | 0 |

| Value | Meaning                  |
|-------|--------------------------|
| 0     | not valid (Not occurred) |
| 1     | valid (occurred)         |

This object is read-only. Although this object is nearly a copy of the object 6425h, it may hold a different value. If, for example, the user wants to switch a controller on, the value TRUE is written to object 6422h. The corresponding bit will be set in object 6425h. However the specified control zone may not be switched on due to an error, so bit 0 in object 6427h will not change to 1.

## OBJECT DESCRIPTION

|             |                |
|-------------|----------------|
| Index       | 6427h          |
| Name        | CO Status Word |
| Object Code | ARRAY          |
| Data Type   | Unsigned16     |
| Category    | Mandatory      |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                  |
|----------------|------------------|
| Sub-Index      | 1h               |
| Description    | CO Status Word 1 |
| Entry Category | Mandatory        |
| Access         | ro               |
| PDO Mapping    | Possible         |
| Value Range    | Unsigned16       |
| Default Value  | No               |

|                |                  |
|----------------|------------------|
| Sub-Index      | 2h               |
| Description    | CO Status Word 2 |
| Entry Category | Optional         |
| Access         | ro               |
| PDO Mapping    | Possible         |
| Value Range    | Unsigned16       |
| Default Value  | No               |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | CO Status Word 199 |
| Entry Category | Optional           |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | 0 - FFFFH          |
| Default Value  | No                 |

**7.2.84 Object x430h: CO Minimum Pulse Time Tmin1**

The physical unit of this object is seconds. The number of decimal digits of this object is defined by object 6434h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x430h                                  |
| Name        | Minimum Pulse Time Tmin1               |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | Minimum Pulse Time Tmin1 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | Minimum Pulse Time Tmin1 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | Minimum Pulse Time Tmin1 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.85 Object x431h: CO Minimum Pulse Time Tmin2**

The physical unit of this object is seconds. The number of decimal digits of this object is defined by object 6434h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x431h                                  |
| Name        | Minimum Pulse Time Tmin2               |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | Minimum Pulse Time Tmin2 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | Minimum Pulse Time Tmin2 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | Minimum Pulse Time Tmin2 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.86 Object x432h: CO Motor Response Time Tm**

Parameter is used for three point stepping controllers. The physical unit of this object is seconds. The number of decimal digits of this object is defined by object 6434h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x432h                                  |
| Name        | CO Motor Response Time Tm              |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | CO Motor Response Time 1 |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | CO Motor Response Time 2 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | data type specific       |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | CO Motor Response Time 199 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

**7.2.87 Object x433h: CO Min. Pulse Length Tpuls**

If the value is set to 0 the pulse length is switched off. The physical unit of this object is seconds. The number of decimal digits of this object is defined by object 6434h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x433h                                  |
| Name        | CO Min Pulse Length Tpuls              |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 1h                    |
| Description    | CO Min Pulse Length 1 |
| Entry Category | Mandatory             |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | data type specific    |
| Default Value  | No                    |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 2h                    |
| Description    | CO Min Pulse Length 2 |
| Entry Category | Optional              |
| Access         | rw                    |
| PDO Mapping    | No                    |
| Value Range    | data type specific    |
| Default Value  | No                    |

to

|                |                         |
|----------------|-------------------------|
| Sub-Index      | C7h                     |
| Description    | CO Min Pulse Length 199 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | data type specific      |
| Default Value  | No                      |

**7.2.88 Object 6434h: CO Physical Unit Pulse Timing**

The object describes the physical unit of the objects beginning from 6430h to 6433h.

**OBJECT DESCRIPTION**

|             |                               |
|-------------|-------------------------------|
| Index       | 6434h                         |
| Name        | CO Physical Unit Pulse Timing |
| Object Code | ARRAY                         |
| Data Type   | Unsigned32                    |
| Category    | Optional                      |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | 1h                              |
| Description    | CO Physical Unit Pulse Timing 1 |
| Entry Category | Mandatory                       |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | Unsigned32                      |
| Default Value  | 0003 0000h                      |

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | 2h                              |
| Description    | CO Physical Unit Pulse Timing 2 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | Unsigned32                      |
| Default Value  | 0003 0000h                      |

to

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | C7h                               |
| Description    | CO Physical Unit Pulse Timing 199 |
| Entry Category | Optional                          |
| Access         | rw                                |
| PDO Mapping    | No                                |
| Value Range    | Unsigned32                        |
| Default Value  | 0003 0000h                        |



**7.2.89 Object 6435h: CO Decimal Digits Pulse Timing**

The object describes the number of decimal digits of the objects beginning from x430h to x433h.

**OBJECT DESCRIPTION**

|             |                      |
|-------------|----------------------|
| Index       | 6434h                |
| Name        | CO Decimal Digits PT |
| Object Code | ARRAY                |
| Data Type   | Unsigned8            |
| Category    | Optional             |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | CO Decimal Digits PT 1 |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 - 15                 |
| Default Value  | No                     |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | CO Decimal Digits PT 2 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | 0 - 15                 |
| Default Value  | No                     |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | CO Decimal Digits PT 199 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 - 15                   |
| Default Value  | No                       |

**7.2.90 Object x440h: CO Neutral Zone 3point Xsh1**

The object describes the neutral zone for a 3-point controller. The physical unit and the number of decimal digits are taken from objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x440h                                  |
| Name        | Neutral zone 3point Xsh1               |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | Neutral zone 3point Xsh1 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | Neutral zone 3point Xsh1 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | Neutral zone 3point Xsh1 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.91 Object x441h: CO Neutral zone 3point Xsh2**

The object describes the neutral zone for a 3-point controller. The physical unit and the number of decimal digits are taken from objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x441h                                  |
| Name        | Neutral zone 3point Xsh2               |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | Neutral zone 3point Xsh2 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | Neutral zone 3point Xsh2 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | Neutral zone 3point Xsh2 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.92 Object x442h: CO Neutral zone 3point stepping Xsh**

The object describes the neutral zone for a 3-point controller. The physical unit and the number of decimal digits are taken from objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x442h                                  |
| Name        | CO Neutral zone 3point stepping Xsh    |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | 1h                                |
| Description    | CO Neutral zone 3point stepping 1 |
| Entry Category | Mandatory                         |
| Access         | rw                                |
| PDO Mapping    | No                                |
| Value Range    | data type specific                |
| Default Value  | No                                |

|                |                                   |
|----------------|-----------------------------------|
| Sub-Index      | 2h                                |
| Description    | CO Neutral zone 3point stepping 2 |
| Entry Category | Optional                          |
| Access         | rw                                |
| PDO Mapping    | No                                |
| Value Range    | data type specific                |
| Default Value  | No                                |

to

|                |                                     |
|----------------|-------------------------------------|
| Sub-Index      | C7h                                 |
| Description    | CO Neutral zone 3point stepping 199 |
| Entry Category | Optional                            |
| Access         | rw                                  |
| PDO Mapping    | No                                  |
| Value Range    | data type specific                  |
| Default Value  | No                                  |

**7.2.93 Object x443h: CO Switching difference of signaller Xsd1**

The object describes the switching difference for a 3-point stepping controller. The physical unit and the number of decimal digits are taken from objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                           |
|-------------|-------------------------------------------|
| Index       | x443h                                     |
| Name        | CO Switching difference of signaller Xsd1 |
| Object Code | ARRAY                                     |
| Data Type   | Float, Integer16, Integer24, Integer32    |
| Category    | Optional                                  |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                             |
|----------------|---------------------------------------------|
| Sub-Index      | 1h                                          |
| Description    | CO Switching difference of signaller Xsd1 1 |
| Entry Category | Mandatory                                   |
| Access         | rw                                          |
| PDO Mapping    | No                                          |
| Value Range    | data type specific                          |
| Default Value  | No                                          |

|                |                                             |
|----------------|---------------------------------------------|
| Sub-Index      | 2h                                          |
| Description    | CO Switching difference of signaller Xsd1 2 |
| Entry Category | Optional                                    |
| Access         | rw                                          |
| PDO Mapping    | No                                          |
| Value Range    | data type specific                          |
| Default Value  | No                                          |

to

|                |                                               |
|----------------|-----------------------------------------------|
| Sub-Index      | C7h                                           |
| Description    | CO Switching difference of signaller Xsd1 199 |
| Entry Category | Optional                                      |
| Access         | rw                                            |
| PDO Mapping    | No                                            |
| Value Range    | data type specific                            |
| Default Value  | No                                            |

**7.2.94 Object x444h: CO Switching difference of additional Contact Xsd2**

The object describes the switching difference for a 3-point stepping controller. The physical unit and the number of decimal digits are taken from objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                                 |
|-------------|-------------------------------------------------|
| Index       | x444h                                           |
| Name        | Switching difference of additional Contact Xsd2 |
| Object Code | ARRAY                                           |
| Data Type   | Float, Integer16, Integer24, Integer32          |
| Category    | Optional                                        |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                                   |
|----------------|---------------------------------------------------|
| Sub-Index      | 1h                                                |
| Description    | Switching difference of additional contact Xsd2 1 |
| Entry Category | Mandatory                                         |
| Access         | rw                                                |
| PDO Mapping    | No                                                |
| Value Range    | data type specific                                |
| Default Value  | No                                                |

|                |                                                   |
|----------------|---------------------------------------------------|
| Sub-Index      | 2h                                                |
| Description    | Switching difference of additional contact Xsd2 2 |
| Entry Category | Optional                                          |
| Access         | rw                                                |
| PDO Mapping    | No                                                |
| Value Range    | data type specific                                |
| Default Value  | No                                                |

to

|                |                                                    |
|----------------|----------------------------------------------------|
| Sub-Index      | C7h                                                |
| Description    | Switching difference of additional contactXsd2 199 |
| Entry Category | Optional                                           |
| Access         | rw                                                 |
| PDO Mapping    | No                                                 |
| Value Range    | data type specific                                 |
| Default Value  | No                                                 |

**7.2.95 Object x445h: CO Trigger point separation of additional contact LW**

The object describes the trigger point for a 3-point stepping controller. The physical unit and the number of decimal digits are taken from objects 6406h and 6407h.

**OBJECT DESCRIPTION**

|             |                                             |
|-------------|---------------------------------------------|
| Index       | x445h                                       |
| Name        | Trigger point separation of add. contact LW |
| Object Code | ARRAY                                       |
| Data Type   | Float, Integer16, Integer24, Integer32      |
| Category    | Optional                                    |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                                               |
|----------------|-----------------------------------------------|
| Sub-Index      | 1h                                            |
| Description    | Trigger point separation of add. contact LW 1 |
| Entry Category | Mandatory                                     |
| Access         | rw                                            |
| PDO Mapping    | No                                            |
| Value Range    | data type specific                            |
| Default Value  | No                                            |

|                |                                               |
|----------------|-----------------------------------------------|
| Sub-Index      | 2h                                            |
| Description    | Trigger point separation of add. contact LW 2 |
| Entry Category | Optional                                      |
| Access         | rw                                            |
| PDO Mapping    | No                                            |
| Value Range    | data type specific                            |
| Default Value  | No                                            |

to

|                |                                                 |
|----------------|-------------------------------------------------|
| Sub-Index      | C7h                                             |
| Description    | Trigger point separation of add. contact LW 199 |
| Entry Category | Optional                                        |
| Access         | rw                                              |
| PDO Mapping    | No                                              |
| Value Range    | data type specific                              |
| Default Value  | No                                              |

**7.2.96 Object x450h: CO Proportional Band Xp1**

The object describes the parameter Xp of the PID-algorithm. The value is given in percent with one digit after the comma (see Object 6410h: CO Effective Controller Output Y for representation as Integer16).

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x440h                                  |
| Name        | CO Proportional Band Xp1               |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | CO Proportional Band Xp1 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | CO Proportional Band Xp1 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | CO Proportional Band Xp1 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |



**7.2.97 Object x451h: CO Proportional Band Xp2**

The object describes the parameter Xp of the PID-algorithm. The value is given in percent with one digit after the comma (see Object 6410h: CO Effective Controller Output Y for representation as Integer16).

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x451h                                  |
| Name        | CO Proportional Band Xp2               |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 1h                         |
| Description    | CO Proportional Band Xp2 1 |
| Entry Category | Mandatory                  |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 2h                         |
| Description    | CO Proportional Band Xp2 2 |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | data type specific         |
| Default Value  | No                         |

to

|                |                              |
|----------------|------------------------------|
| Sub-Index      | C7h                          |
| Description    | CO Proportional Band Xp2 199 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

**7.2.98 Object x452h: CO Integral Action Time Tn1**

The object describes the parameter Tn of the PID-algorithm. The physical unit and the number of decimal digits is given by objects 6458h and 6459h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x452h                                  |
| Name        | CO Integral Action Time Tn1            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | CO Integral Action Time Tn1 1 |
| Entry Category | Mandatory                     |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | CO Integral Action Time Tn1 2 |
| Entry Category | Optional                      |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | CO Integral Action Time Tn1 199 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | data type specific              |
| Default Value  | No                              |

**7.2.99 Object x453h: CO Integral Action Time Tn2**

The object describes the parameter Tn of the PID-algorithm. The physical unit and the number of decimal digits is given by objects 6458h and 6459h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x453h                                  |
| Name        | CO Integral Action Time Tn2            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | CO Integral Action Time Tn2 1 |
| Entry Category | Mandatory                     |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | CO Integral Action Time Tn2 2 |
| Entry Category | Optional                      |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | CO Integral Action Time Tn2 199 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | data type specific              |
| Default Value  | No                              |

**7.2.100 Object x454h: CO Derivate Action Time Tv1**

The object describes the parameter Tv of the PID-algorithm. The physical unit and the number of decimal digits is given by objects 6458h and 6459h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x454h                                  |
| Name        | CO Derivate Action Time Tv1            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | CO Derivate Action Time Tv1 1 |
| Entry Category | Mandatory                     |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | CO Derivate Action Time Tv1 2 |
| Entry Category | Optional                      |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | CO Derivate Action Time Tv1 199 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | data type specific              |
| Default Value  | No                              |

**7.2.101 Object x455h: CO Derivate Action Time Tv2**

The object describes the parameter Tv of the PID-algorithm. The physical unit and the number of decimal digits is given by objects 6458h and 6459h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x455h                                  |
| Name        | CO Derivate Action Time Tv2            |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 1h                            |
| Description    | CO Derivate Action Time Tv2 1 |
| Entry Category | Mandatory                     |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

|                |                               |
|----------------|-------------------------------|
| Sub-Index      | 2h                            |
| Description    | CO Derivate Action Time Tv2 2 |
| Entry Category | Optional                      |
| Access         | rw                            |
| PDO Mapping    | No                            |
| Value Range    | data type specific            |
| Default Value  | No                            |

to

|                |                                 |
|----------------|---------------------------------|
| Sub-Index      | C7h                             |
| Description    | CO Derivate Action Time Tv2 199 |
| Entry Category | Optional                        |
| Access         | rw                              |
| PDO Mapping    | No                              |
| Value Range    | data type specific              |
| Default Value  | No                              |

**7.2.102 Object x456h: CO Cycle Time T1**

The object describes the cycle time of the PID-algorithm. The physical unit and the number of decimal digits is given by objects 6458h and 6459h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x456h                                  |
| Name        | CO Cycle Time T1                       |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | CO Cycle Time T1 1 |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | CO Cycle Time T1 2 |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | No                 |

to

|                |                      |
|----------------|----------------------|
| Sub-Index      | 199                  |
| Description    | CO Cycle Time T1 199 |
| Entry Category | Optional             |
| Access         | rw                   |
| PDO Mapping    | No                   |
| Value Range    | data type specific   |
| Default Value  | No                   |

**7.2.103 Object x457h: CO Cycle Time T2**

The object describes the cycle time of the PID-algorithm. The physical unit and the number of decimal digits is given by objects 6458h and 6459h.

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x457h                                  |
| Name        | CO Cycle Time T2                       |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | CO Cycle Time T2 1 |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | No                 |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | CO Cycle Time T2 2 |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | No                 |

to

|                |                      |
|----------------|----------------------|
| Sub-Index      | C7h                  |
| Description    | CO Cycle Time T2 199 |
| Entry Category | Optional             |
| Access         | rw                   |
| PDO Mapping    | No                   |
| Value Range    | data type specific   |
| Default Value  | No                   |

**7.2.104 Object 6458h: CO Physical Unit PID Timing**

The object describes the physical unit of the objects beginning from x450h to x457h (see Object 6131h: AI Physical unit PV).

**OBJECT DESCRIPTION**

|             |                      |
|-------------|----------------------|
| Index       | 6458h                |
| Name        | CO Physical Unit PID |
| Object Code | ARRAY                |
| Data Type   | Unsigned32           |
| Category    | Optional             |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 1h                     |
| Description    | CO Physical Unit PID 1 |
| Entry Category | Mandatory              |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | Unsigned32             |
| Default Value  | 0003 0000h             |

|                |                        |
|----------------|------------------------|
| Sub-Index      | 2h                     |
| Description    | CO Physical Unit PID 2 |
| Entry Category | Optional               |
| Access         | rw                     |
| PDO Mapping    | No                     |
| Value Range    | Unsigned32             |
| Default Value  | 0003 0000h             |

to

|                |                          |
|----------------|--------------------------|
| Sub-Index      | C7h                      |
| Description    | CO Physical Unit PID 199 |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | Unsigned32               |
| Default Value  | 0003 0000h               |



**7.2.105 Object 6459h: CO Decimal Digits PID Timing**

The object describes the number of decimal digits of the objects beginning from x450h to x457h.

**OBJECT DESCRIPTION**

|             |                       |
|-------------|-----------------------|
| Index       | 6459h                 |
| Name        | CO Decimal Digits PID |
| Object Code | ARRAY                 |
| Data Type   | Unsigned8             |
| Category    | Optional              |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | CO Decimal Digits PID 1 |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | 0 - 15                  |
| Default Value  | No                      |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | CO Decimal Digits PID 2 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | No                      |
| Value Range    | 0 - 15                  |
| Default Value  | No                      |

to

|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | CO Decimal Digits PID 199 |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | 0 - 15                    |
| Default Value  | No                        |

**7.2.106 Object x500h: AL 1 Input value**

This object represents the process value, which is the input to the alarm function block. It is scaled in the physical unit of process values (e.g. degrees centigrade, kg, kN, mm etc.).

## OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x500h                                  |
| Name        | <b>AL 1 Input value</b>                |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Mandatory                              |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                           |
|----------------|---------------------------|
| Sub-Index      | 1h                        |
| Description    | <b>AL 1 Input value 1</b> |
| Entry Category | Mandatory                 |
| Access         | rw                        |
| PDO Mapping    | Possible                  |
| Value Range    | data type specific        |
| Default Value  | No                        |

|                |                           |
|----------------|---------------------------|
| Sub-Index      | 2h                        |
| Description    | <b>AL 1 Input value 2</b> |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | Possible                  |
| Value Range    | data type specific        |
| Default Value  | No                        |

to

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | C7h                         |
| Description    | <b>AL 1 Input value 199</b> |
| Entry Category | Optional                    |
| Access         | rw                          |
| PDO Mapping    | Possible                    |
| Value Range    | data type specific          |
| Default Value  | No                          |

**7.2.107 Object 6503h: AL 1 Link input**

This object describes the source for the AL Input Value if it lies within another functional block of the same device. The object defined by index and sub-index is directly linked to the AL Input Value.

**OBJECT DESCRIPTION**

|             |                        |
|-------------|------------------------|
| Index       | 6503h                  |
| Name        | <b>AL 1 Link input</b> |
| Object Code | RECORD                 |
| Data Type   | PDOmapping             |
| Category    | Optional               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | <b>AL 1 Link input 1</b> |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | Unsigned32               |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | <b>AL 1 Link input 2</b> |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | Unsigned32               |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | <b>AL 1 Link input 199</b> |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | Unsigned32                 |
| Default Value  | No                         |

**7.2.108 Object x504h: AL 1 Alternate input value**

This object represents the process value, which is the alternate input to the alarm function block. This alternate input is only used if functionality needs two input values.

It is scaled in the physical unit of process values (e.g. degrees centigrade, kg, kN, mm etc.).

**OBJECT DESCRIPTION**

|             |                                        |
|-------------|----------------------------------------|
| Index       | x504h                                  |
| Name        | AL 1 Alternate input value             |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                              |
|----------------|------------------------------|
| Sub-Index      | 1h                           |
| Description    | AL 1 Alternate input value 1 |
| Entry Category | Mandatory                    |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

|                |                              |
|----------------|------------------------------|
| Sub-Index      | 2h                           |
| Description    | AL 1 Alternate input value 2 |
| Entry Category | Optional                     |
| Access         | rw                           |
| PDO Mapping    | No                           |
| Value Range    | data type specific           |
| Default Value  | No                           |

to

|                |                                |
|----------------|--------------------------------|
| Sub-Index      | C7h                            |
| Description    | AL 1 Alternate input value 199 |
| Entry Category | Optional                       |
| Access         | rw                             |
| PDO Mapping    | No                             |
| Value Range    | data type specific             |
| Default Value  | No                             |

**7.2.109 Object 6505h: AL 1 Link alternate input**

This object describes the source for the AL Input Value if it lies within another functional block of the same device. The object defined by index and sub-index is directly linked to the AL Input Value.

**OBJECT DESCRIPTION**

|             |                                  |
|-------------|----------------------------------|
| Index       | 6505H                            |
| Name        | <b>AL 1 Link alternate input</b> |
| Object Code | RECORD                           |
| Data Type   | PDOmapping                       |
| Category    | Optional                         |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 – C7h           |
| Default Value  | No                |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 1h                                 |
| Description    | <b>AL 1 Link alternate input 1</b> |
| Entry Category | Mandatory                          |
| Access         | rw                                 |
| PDO Mapping    | No                                 |
| Value Range    | 0 – 9h                             |
| Default Value  | No                                 |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 2h                                 |
| Description    | <b>AL 1 Link alternate input 2</b> |
| Entry Category | Optional                           |
| Access         | rw                                 |
| PDO Mapping    | No                                 |
| Value Range    | 0 – 9h                             |
| Default Value  | No                                 |

to

|                |                                      |
|----------------|--------------------------------------|
| Sub-Index      | C7h                                  |
| Description    | <b>AL 1 Link alternate input 199</b> |
| Entry Category | Optional                             |
| Access         | rw                                   |
| PDO Mapping    | No                                   |
| Value Range    | 0 – 9h                               |
| Default Value  | No                                   |

**7.2.110 Object 6506h: AL 1 Fault mode**

A value of 0 specifies that the state of output AL 1 is not changed in case of a fault condition. If value is 1 then AL 1 Fault State is output of AL 1 state if there is a fault condition.

**OBJECT DESCRIPTION**

|             |                        |
|-------------|------------------------|
| Index       | 6506h                  |
| Name        | <b>AL 1 Fault mode</b> |
| Object Code | ARRAY                  |
| Data Type   | Unsigned8              |
| Category    | Optional               |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | <b>AL 1 Fault mode 1</b> |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 – 1h                   |
| Default Value  | 1h                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | <b>AL 1 Fault mode 2</b> |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | No                       |
| Value Range    | 0 – 1h                   |
| Default Value  | 1h                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | 199                        |
| Description    | <b>AL 1 Fault mode 199</b> |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | No                         |
| Value Range    | 0 – 1h                     |
| Default Value  | 1h                         |

**7.2.111 Object 6507h: AL 1 Fault state**

State of Alarm output, if there is a fault condition and AL 1 Fault Mode is 1.

**OBJECT DESCRIPTION**

|             |                         |
|-------------|-------------------------|
| Index       | 6507h                   |
| Name        | <b>AL 1 Fault state</b> |
| Object Code | ARRAY                   |
| Data Type   | Boolean                 |
| Category    | Optional                |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                           |
|----------------|---------------------------|
| Sub-Index      | 1h                        |
| Description    | <b>AL 1 Fault state 1</b> |
| Entry Category | Mandatory                 |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | Boolean                   |
| Default Value  | FALSE                     |

|                |                           |
|----------------|---------------------------|
| Sub-Index      | 2h                        |
| Description    | <b>AL 1 Fault state 2</b> |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | No                        |
| Value Range    | Boolean                   |
| Default Value  | FALSE                     |

to

|                |                             |
|----------------|-----------------------------|
| Sub-Index      | C7h                         |
| Description    | <b>AL 1 Fault state 199</b> |
| Entry Category | Optional                    |
| Access         | rw                          |
| PDO Mapping    | No                          |
| Value Range    | Boolean                     |
| Default Value  | FALSE                       |

**7.2.112 Object 6508h: AL 1 Type**

This object specifies the alarm type.

| Value | Meaning                                    |
|-------|--------------------------------------------|
| 0h    | No Alarm Function                          |
| 1h    | Sensor Fault                               |
| 2h    | above or equal                             |
| 3h    | Below                                      |
| 4h    | Difference (input - alternate input) above |
| 5h    | Difference below                           |
| 6h    | in window (including borders)              |
| 7h    | out of window (excluding borders)          |
| 8h    | reserved                                   |
| :::   | :::                                        |
| 11h   | reserved                                   |
| 12h   | above with starting condition              |
| 13h   | below with starting condition              |
| 14h   | Difference above with starting condition   |
| 15h   | Difference below with starting condition   |
| 16h   | in window with starting condition          |
| 17h   | out of window with starting condition      |
| 18h   | reserved                                   |
| :::   | :::                                        |
| FFh   | reserved                                   |

**OBJECT DESCRIPTION**

|             |           |
|-------------|-----------|
| Index       | 6508h     |
| Name        | AL 1 Type |
| Object Code | ARRAY     |
| Data Type   | Unsigned8 |
| Category    | Optional  |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |             |
|----------------|-------------|
| Sub-Index      | 1h          |
| Description    | AL 1 Type 1 |
| Entry Category | Mandatory   |
| Access         | rw          |
| PDO Mapping    | No          |
| Value Range    | Unsigned8   |
| Default Value  | 0           |



|                |             |
|----------------|-------------|
| Sub-Index      | 2h          |
| Description    | AL 1 Type 2 |
| Entry Category | Optional    |
| Access         | rw          |
| PDO Mapping    | No          |
| Value Range    | Unsigned8   |
| Default Value  | 0           |

to

|                |               |
|----------------|---------------|
| Sub-Index      | C7h           |
| Description    | AL 1 Type 199 |
| Entry Category | Optional      |
| Access         | rw            |
| PDO Mapping    | No            |
| Value Range    | Unsigned8     |
| Default Value  | 0             |

### 7.2.113 Object 6509h: AL 1 Action

This object defines which actions are performed, if the alarm state is changing.

Objects 6519h, 6529h ... 65F9h are similar defined.

MSB

LSB

|          |   |                                                     |                                                     |                   |                              |
|----------|---|-----------------------------------------------------|-----------------------------------------------------|-------------------|------------------------------|
| reserved |   | Interrupt 1-0 change<br>(send 3 <sup>rd</sup> TPDO) | Interrupt 0-1 change<br>(send 3 <sup>rd</sup> TPDO) | Internal<br>Alarm | Send<br>Emergency<br>Message |
| 15       | 4 | 3                                                   | 2                                                   | 1                 | 0                            |

| Value | Meaning   |
|-------|-----------|
| 0     | not valid |
| 1     | valid     |

#### OBJECT DESCRIPTION

|             |             |
|-------------|-------------|
| Index       | 6509h       |
| Name        | AL 1 Action |
| Object Code | ARRAY       |
| Data Type   | Unsigned8   |
| Category    | Optional    |

#### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |               |
|----------------|---------------|
| Sub-Index      | 1h            |
| Description    | AL 1 Action 1 |
| Entry Category | Mandatory     |
| Access         | rw            |
| PDO Mapping    | No            |
| Value Range    | Unsigned8     |
| Default Value  | 0             |

|                |               |
|----------------|---------------|
| Sub-Index      | 2h            |
| Description    | AL 1 Action 2 |
| Entry Category | Optional      |
| Access         | rw            |
| PDO Mapping    | No            |
| Value Range    | Unsigned8     |
| Default Value  | 0             |

to

|                |                 |
|----------------|-----------------|
| Sub-Index      | C7h             |
| Description    | AL 1 Action 199 |
| Entry Category | Optional        |
| Access         | rw              |
| PDO Mapping    | No              |
| Value Range    | Unsigned8       |
| Default Value  | 0               |

#### 7.2.114 Object x50Ah: AL 1 Level

This object represents the level, which the AL Input Value is compared with. It is scaled in the physical unit of process values (e.g. degrees centigrade, kg, kN, mm etc.). If alarm type is "window" then this object represents the lower limit of the window.

Objects x51Ah, x52Ah ... x5FAh are defined similar.

##### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x50Ah                                  |
| Name        | AL 1 Level                             |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Mandatory                              |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AL 1 Level 1       |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AL 1 Level 2       |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AL 1 Level 199     |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 0                  |

#### 7.2.115 Object x50Bh: AL 1 Hysteresis

This object represents the hysteresis, which is used for the comparison of AL Input Value and AL Level. It helps the AL State not to alter too fast when AL Input Values are noisy. Hysteresis is scaled in the physical unit of process values (e.g. degrees centigrade, kg, kN, mm etc.). If alarm type is "window" then this object specifies the width of the window.

Objects x51Bh, x52Bh ... x5FBh are defined similar.

##### OBJECT DESCRIPTION

|             |                                        |
|-------------|----------------------------------------|
| Index       | x50Bh                                  |
| Name        | AL 1 Hysteresis                        |
| Object Code | ARRAY                                  |
| Data Type   | Float, Integer16, Integer24, Integer32 |
| Category    | Optional                               |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 1h                 |
| Description    | AL 1 Hysteresis 1  |
| Entry Category | Mandatory          |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 1                  |

|                |                    |
|----------------|--------------------|
| Sub-Index      | 2h                 |
| Description    | AL 1 Hysteresis 2  |
| Entry Category | Optional           |
| Access         | rw                 |
| PDO Mapping    | No                 |
| Value Range    | data type specific |
| Default Value  | 1                  |

to

|                |                     |
|----------------|---------------------|
| Sub-Index      | C7h                 |
| Description    | AL 1 Hysteresis 199 |
| Entry Category | Optional            |
| Access         | rw                  |
| PDO Mapping    | No                  |
| Value Range    | data type specific  |
| Default Value  | 1                   |

#### 7.2.116 Object 650Ch: AL 1 Group

Alarm Outputs can be grouped together, that multiple alarms are grouped to a single AL Group State bit. A maximum of 8 groups can be defined. A value of 0 means, that the alarm state is not tied to a alarm group.

Objects 651Ch, 652Ch ... 65FCh are defined similar.

##### OBJECT DESCRIPTION

|             |            |
|-------------|------------|
| Index       | 650Ch      |
| Name        | AL 1 Group |
| Object Code | ARRAY      |
| Data Type   | Unsigned8  |
| Category    | Optional   |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |              |
|----------------|--------------|
| Sub-Index      | 1h           |
| Description    | AL 1 Group 1 |
| Entry Category | Mandatory    |
| Access         | rw           |
| PDO Mapping    | No           |
| Value Range    | 0 - 8        |
| Default Value  | 0            |

|                |              |
|----------------|--------------|
| Sub-Index      | 2h           |
| Description    | AL 1 Group 2 |
| Entry Category | Optional     |
| Access         | rw           |
| PDO Mapping    | No           |
| Value Range    | 0 - 8        |
| Default Value  | 0            |

to

|                |                |
|----------------|----------------|
| Sub-Index      | C7h            |
| Description    | AL 1 Group 199 |
| Entry Category | Optional       |
| Access         | rw             |
| PDO Mapping    | No             |
| Value Range    | 0 - 8          |
| Default Value  | 0              |

#### 7.2.117 Object 650Dh: AL 1 State

Alarm state represents the result of the alarm comparison.

Objects 651Dh, 652Dh ... 65FDh are defined similar.

##### OBJECT DESCRIPTION

|             |            |
|-------------|------------|
| Index       | 650Dh      |
| Name        | AL 1 State |
| Object Code | ARRAY      |
| Data Type   | Boolean    |
| Category    | Optional   |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |              |
|----------------|--------------|
| Sub-Index      | 1h           |
| Description    | AL 1 State 1 |
| Entry Category | Mandatory    |
| Access         | ro           |
| PDO Mapping    | Possible     |
| Value Range    | Boolean      |
| Default Value  | No           |

|                |              |
|----------------|--------------|
| Sub-Index      | 2h           |
| Description    | AL 1 State 2 |
| Entry Category | Optional     |
| Access         | ro           |
| PDO Mapping    | Possible     |
| Value Range    | Boolean      |
| Default Value  | No           |

to

|                |                |
|----------------|----------------|
| Sub-Index      | C7h            |
| Description    | AL 1 State 199 |
| Entry Category | Optional       |
| Access         | ro             |
| PDO Mapping    | Possible       |
| Value Range    | Boolean        |
| Default Value  | No             |

#### 7.2.118 Object 650Eh: AL 1 Reset

This write-only object resets the alarm function block, which is important, if hysteresis or starting conditions are defined.

Objects 651Eh, 652Eh ... 65FEh are defined similar.

##### OBJECT DESCRIPTION

|             |            |
|-------------|------------|
| Index       | 650Eh      |
| Name        | AL 1 Reset |
| Object Code | ARRAY      |
| Data Type   | Boolean    |
| Category    | Optional   |

##### ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |              |
|----------------|--------------|
| Sub-Index      | 1h           |
| Description    | AL 1 Reset 1 |
| Entry Category | Mandatory    |
| Access         | w0           |
| PDO Mapping    | No           |
| Value Range    | Boolean      |
| Default Value  | No           |

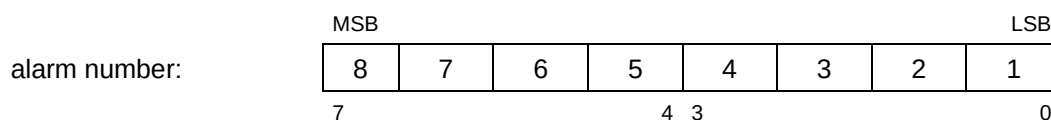
|                |              |
|----------------|--------------|
| Sub-Index      | 2h           |
| Description    | AL 1 Reset 2 |
| Entry Category | Optional     |
| Access         | w0           |
| PDO Mapping    | No           |
| Value Range    | Boolean      |
| Default Value  | No           |

to

|                |                |
|----------------|----------------|
| Sub-Index      | C7h            |
| Description    | AL 1 Reset 199 |
| Entry Category | Optional       |
| Access         | w0             |
| PDO Mapping    | No             |
| Value Range    | Boolean        |
| Default Value  | No             |

#### 7.2.119 Object 6600h: AL 1..8 State

Alarm state represents the result of the alarm comparison of alarm number 1 to 8.



| Value | Meaning      |
|-------|--------------|
| 0     | not occurred |
| 1     | occurred     |

#### OBJECT DESCRIPTION

|             |               |
|-------------|---------------|
| Index       | 6600H         |
| Name        | AL 1..8 State |
| Object Code | ARRAY         |
| Data Type   | Unsigned8     |
| Category    | Mandatory     |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                 |
|----------------|-----------------|
| Sub-Index      | 1h              |
| Description    | AL 1..8 State 1 |
| Entry Category | Mandatory       |
| Access         | ro              |
| PDO Mapping    | Possible        |
| Value Range    | Unsigned8       |
| Default Value  | No              |

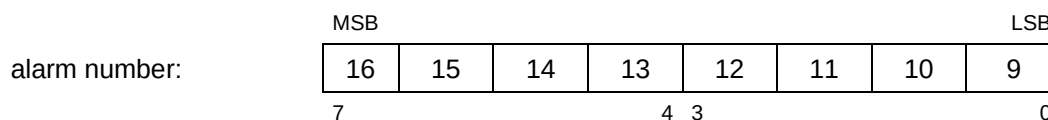
|                |                 |
|----------------|-----------------|
| Sub-Index      | 2h              |
| Description    | AL 1..8 State 2 |
| Entry Category | Optional        |
| Access         | ro              |
| PDO Mapping    | Possible        |
| Value Range    | Unsigned8       |
| Default Value  | No              |

to

|                |                   |
|----------------|-------------------|
| Sub-Index      | C7h               |
| Description    | AL 1..8 State 199 |
| Entry Category | Optional          |
| Access         | ro                |
| PDO Mapping    | Possible          |
| Value Range    | Unsigned8         |
| Default Value  | No                |

**7.2.120 Object 6601h: AL 9..16 State**

Alarm state represents the result of the alarm comparison of alarm number 9 to 16.



| Value | Meaning      |
|-------|--------------|
| 0     | not occurred |
| 1     | occurred     |



## OBJECT DESCRIPTION

|             |                |
|-------------|----------------|
| Index       | 6601h          |
| Name        | AL 9..16 State |
| Object Code | ARRAY          |
| Data Type   | Unsigned8      |
| Category    | Optional       |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | Number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                  |
|----------------|------------------|
| Sub-Index      | 1h               |
| Description    | AL 9..16 State 1 |
| Entry Category | Mandatory        |
| Access         | ro               |
| PDO Mapping    | Possible         |
| Value Range    | Unsigned8        |
| Default Value  | No               |

|                |                  |
|----------------|------------------|
| Sub-Index      | 2h               |
| Description    | AL 9..16 State 2 |
| Entry Category | Optional         |
| Access         | ro               |
| PDO Mapping    | Possible         |
| Value Range    | Unsigned8        |
| Default Value  | No               |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AL 9..16 State 199 |
| Entry Category | Optional           |
| Access         | ro                 |
| PDO Mapping    | Possible           |
| Value Range    | Unsigned8          |
| Default Value  | No                 |

**7.2.121 Object 6602h: AL General state**

Logical OR-result of all alarm states of all alarm channels.

**OBJECT DESCRIPTION**

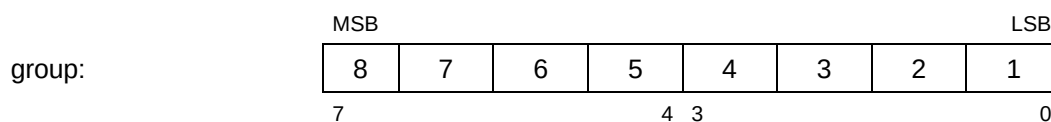
|             |                  |
|-------------|------------------|
| Index       | 6602             |
| Name        | AL General state |
| Object Code | VAR              |
| Data Type   | Boolean          |
| Category    | Optional         |

**ENTRY DESCRIPTION**

|               |          |
|---------------|----------|
| Access        | ro       |
| PDO Mapping   | Possible |
| Value Range   | Boolean  |
| Default Value | No       |

**7.2.122 Object 6603h: AL Group 1..8 State**

A single bit in the group alarm state is set to one if at least one of the alarms tied to this group is true.



| Value | Meaning      |
|-------|--------------|
| 0     | not occurred |
| 1     | occurred     |

**OBJECT DESCRIPTION**

|             |                     |
|-------------|---------------------|
| Index       | 6603h               |
| Name        | AL Group 1..8 State |
| Object Code | VAR                 |
| Data Type   | Unsigned8           |
| Category    | Optional            |

**ENTRY DESCRIPTION**

|               |           |
|---------------|-----------|
| Access        | ro        |
| PDO Mapping   | Possible  |
| Value Range   | Unsigned8 |
| Default Value | No        |

**7.2.123 Object 6610h: AL General Reset**

This write only object resets all alarms of all channels within the unit which is important, if hysteresis or starting conditions are defined.

**OBJECT DESCRIPTION**

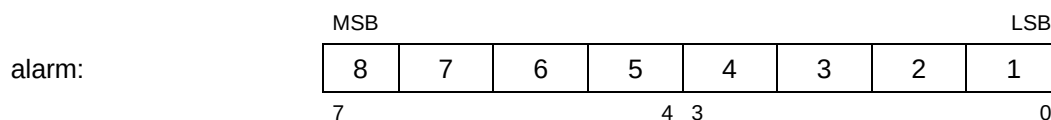
|             |                  |
|-------------|------------------|
| Index       | 6610h            |
| Name        | AL General Reset |
| Object Code | VAR              |
| Data Type   | Boolean          |
| Category    | Optional         |

**OBJECT DESCRIPTION**

|               |          |
|---------------|----------|
| Access        | wo       |
| PDO Mapping   | Possible |
| Value Range   | Boolean  |
| Default Value | No       |

**7.2.124 Object 6611h: AL 1..8 Reset**

Alarm reset bits of alarm 1 to 8 grouped together within one byte.



| Value | Meaning              |
|-------|----------------------|
| 0     | do not perform reset |
| 1     | perform reset        |

**OBJECT DESCRIPTION**

|             |               |
|-------------|---------------|
| Index       | 6611h         |
| Name        | AL 1..8 Reset |
| Object Code | ARRAY         |
| Data Type   | Unsigned8     |
| Category    | Optional      |

**ENTRY DESCRIPTION**

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                 |
|----------------|-----------------|
| Sub-Index      | 1h              |
| Description    | AL 1..8 Reset 1 |
| Entry Category | Mandatory       |
| Access         | w0              |
| PDO Mapping    | Possible        |
| Value Range    | Unsigned8       |
| Default Value  | No              |

|                |                 |
|----------------|-----------------|
| Sub-Index      | 2h              |
| Description    | AL 1..8 Reset 2 |
| Entry Category | Optional        |
| Access         | w0              |
| PDO Mapping    | Possible        |
| Value Range    | Unsigned8       |
| Default Value  | No              |

to

|                |                   |
|----------------|-------------------|
| Sub-Index      | C7h               |
| Description    | AL 1..8 Reset 199 |
| Entry Category | Optional          |
| Access         | w0                |
| PDO Mapping    | Possible          |
| Value Range    | Unsigned8         |
| Default Value  | No                |

**7.2.125 Object 6612h: AL 9..16 Reset**

Alarm reset bits of alarm 9 to 16 grouped together within in byte.

|        |     |    |    |    |    |    |    |     |
|--------|-----|----|----|----|----|----|----|-----|
|        | MSB |    |    |    |    |    |    | LSB |
| alarm: | 16  | 15 | 14 | 13 | 12 | 11 | 10 | 9   |
|        | 7   |    |    | 4  | 3  |    |    | 0   |

| Value | Meaning              |
|-------|----------------------|
| 0     | do not perform reset |
| 1     | perform reset        |

**OBJECT DESCRIPTION**

|             |                |
|-------------|----------------|
| Index       | 6612h          |
| Name        | AL 9..16 Reset |
| Object Code | ARRAY          |
| Data Type   | Unsigned8      |
| Category    | Optional       |

## ENTRY DESCRIPTION

|                |                   |
|----------------|-------------------|
| Sub-Index      | 0h                |
| Description    | number of entries |
| Entry Category | Mandatory         |
| Access         | ro                |
| PDO Mapping    | No                |
| Value Range    | 1 - C7h           |
| Default Value  | No                |

|                |                  |
|----------------|------------------|
| Sub-Index      | 1h               |
| Description    | AL 9..16 Reset 1 |
| Entry Category | Mandatory        |
| Access         | wo               |
| PDO Mapping    | Possible         |
| Value Range    | Unsigned8        |
| Default Value  | No               |

|                |                  |
|----------------|------------------|
| Sub-Index      | 2h               |
| Description    | AL 9..16 Reset 2 |
| Entry Category | Optional         |
| Access         | wo               |
| PDO Mapping    | Possible         |
| Value Range    | Unsigned8        |
| Default Value  | No               |

to

|                |                    |
|----------------|--------------------|
| Sub-Index      | C7h                |
| Description    | AL 9..16 Reset 199 |
| Entry Category | Optional           |
| Access         | wo                 |
| PDO Mapping    | Possible           |
| Value Range    | Unsigned8          |
| Default Value  | No                 |

**7.2.126 Object 6F20h: Life counter**

The life counter is incremented in a manufacturer specific way to make sure for a receiver of PDO data that the system is still alive and is producing new values.

## OBJECT DESCRIPTION

|             |                     |
|-------------|---------------------|
| Index       | 6F20h               |
| Name        | <b>Life counter</b> |
| Object Code | ARRAY               |
| Data Type   | Unsigned8           |
| Category    | Optional            |

## ENTRY DESCRIPTION

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 0h                      |
| Description    | number of valid entries |
| Entry Category | Mandatory               |
| Access         | ro                      |
| PDO Mapping    | No                      |
| Value Range    | 0 - C7h                 |
| Default Value  | No                      |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 1h                    |
| Description    | <b>Life counter 1</b> |
| Entry Category | Mandatory             |
| Access         | ro                    |
| PDO Mapping    | Possible              |
| Value Range    | Unsigned8             |
| Default Value  | 0                     |

|                |                       |
|----------------|-----------------------|
| Sub-Index      | 2h                    |
| Description    | <b>Life counter 2</b> |
| Entry Category | Optional              |
| Access         | ro                    |
| PDO Mapping    | Possible              |
| Value Range    | Unsigned8             |
| Default Value  | 0                     |

to

|                |                         |
|----------------|-------------------------|
| Sub-Index      | C7h                     |
| Description    | <b>Life counter 199</b> |
| Entry Category | Optional                |
| Access         | ro                      |
| PDO Mapping    | Possible                |
| Value Range    | Unsigned8               |
| Default Value  | 0                       |

**7.2.127 Object 6F30h: Receive PDO check**

This object defines check-bytes for receive PDOs 1 to n. If mapped into a receive PDO, the check-byte is copied from the receive PDO to the data object each time the PDO is received and can be evaluated by the application. The algorithm for calculating the check-byte is manufacturer specific.

## OBJECT DESCRIPTION

|             |                          |
|-------------|--------------------------|
| Index       | 6F30h                    |
| Name        | <b>Receive PDO check</b> |
| Object Code | ARRAY                    |
| Data Type   | Unsigned8                |
| Category    | Optional                 |

## ENTRY DESCRIPTION

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 0h                      |
| Description    | number of valid entries |
| Entry Category | Mandatory               |
| Access         | ro                      |
| PDO Mapping    | No                      |
| Value Range    | 0 – 10h                 |
| Default Value  | No                      |

|                |              |
|----------------|--------------|
| Sub-Index      | 1h           |
| Description    | rPDO 1 Check |
| Entry Category | Mandatory    |
| Access         | wo           |
| PDO Mapping    | Possible     |
| Value Range    | Unsigned8    |
| Default Value  | No           |

|                |              |
|----------------|--------------|
| Sub-Index      | 2h           |
| Description    | rPDO 2 Check |
| Entry Category | Optional     |
| Access         | wo           |
| PDO Mapping    | Possible     |
| Value Range    | Unsigned8    |
| Default Value  | No           |

to

|                |               |
|----------------|---------------|
| Sub-Index      | 10h           |
| Description    | rPDO 16 Check |
| Entry Category | Optional      |
| Access         | wo            |
| PDO Mapping    | Possible      |
| Value Range    | Unsigned8     |
| Default Value  | No            |

**7.2.128 Object 6F31h: Transmit PDO check**

This object defines check-bytes for transmit PDOs 1 to n. If mapped into a transmit PDO, the check-byte is calculated and copied to the transmit PDO each time the PDO is transmitted. The algorithm for calculating the check-byte is manufacturer specific.

## OBJECT DESCRIPTION

|             |                           |
|-------------|---------------------------|
| Index       | 6F30h                     |
| Name        | <b>Transmit PDO check</b> |
| Object Code | ARRAY                     |
| Data Type   | Unsigned8                 |
| Category    | Optional                  |

## ENTRY DESCRIPTION

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 0h                      |
| Description    | number of valid entries |
| Entry Category | Mandatory               |
| Access         | ro                      |
| PDO Mapping    | No                      |
| Value Range    | 0 – 10h                 |
| Default Value  | No                      |

|                |              |
|----------------|--------------|
| Sub-Index      | 1h           |
| Description    | tPDO 1 Check |
| Entry Category | Mandatory    |
| Access         | ro           |
| PDO Mapping    | Possible     |
| Value Range    | Unsigned8    |
| Default Value  | No           |

|                |              |
|----------------|--------------|
| Sub-Index      | 2h           |
| Description    | tPDO 2 Check |
| Entry Category | Optional     |
| Access         | ro           |
| PDO Mapping    | Possible     |
| Value Range    | Unsigned8    |
| Default Value  | No           |

to

|                |               |
|----------------|---------------|
| Sub-Index      | 10h           |
| Description    | tPDO 16 Check |
| Entry Category | Optional      |
| Access         | ro            |
| PDO Mapping    | Possible      |
| Value Range    | Unsigned8     |
| Default Value  | No            |

**7.2.129 Object xF50h: Received PV**

General input process value, which is used as input for analogue output (output process value), controller (effective current value) or alarm (input value for alarm 1..16) function block depending on type of module.

## OBJECT DESCRIPTION

|             |                    |
|-------------|--------------------|
| Index       | xF50h              |
| Name        | Received PV        |
| Object Code | ARRAY              |
| Data Type   | specified by index |
| Category    | Mandatory          |



## ENTRY DESCRIPTION

|                |                    |
|----------------|--------------------|
| Sub-Index      | 0h                 |
| Description    | number of channels |
| Entry Category | Mandatory          |
| Access         | ro                 |
| PDO Mapping    | No                 |
| Value Range    | 1 - C7h            |
| Default Value  | No                 |

|                |               |
|----------------|---------------|
| Sub-Index      | 1h            |
| Description    | Received PV 1 |
| Entry Category | Mandatory     |
| Access         | rw            |
| PDO Mapping    | Possible      |
| Value Range    | Unsigned8     |
| Default Value  | No            |

|                |               |
|----------------|---------------|
| Sub-Index      | 2h            |
| Description    | Received PV 2 |
| Entry Category | Optional      |
| Access         | rw            |
| PDO Mapping    | Possible      |
| Value Range    | Unsigned8     |
| Default Value  | No            |

to

|                |                 |
|----------------|-----------------|
| Sub-Index      | C7h             |
| Description    | Received PV 199 |
| Entry Category | Optional        |
| Access         | rw              |
| PDO Mapping    | Possible        |
| Value Range    | Unsigned8       |
| Default Value  | No              |

**7.2.130 Object xF51h: Alternate received PV**

Alternate input process value, which is used as input for alarm (alternate input value for alarm 1..16) function block.

## OBJECT DESCRIPTION

|             |                       |
|-------------|-----------------------|
| Index       | xF51h                 |
| Name        | Alternate received PV |
| Object Code | ARRAY                 |
| Data Type   | specified by index    |
| Category    | Optional              |

## ENTRY DESCRIPTION

|                |                    |
|----------------|--------------------|
| Sub-Index      | 0h                 |
| Description    | number of channels |
| Entry Category | Mandatory          |
| Access         | ro                 |
| PDO Mapping    | No                 |
| Value Range    | 1 - C7h            |
| Default Value  | No                 |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 1h                      |
| Description    | Alternate received PV 1 |
| Entry Category | Mandatory               |
| Access         | rw                      |
| PDO Mapping    | Possible                |
| Value Range    | Unsigned8               |
| Default Value  | No                      |

|                |                         |
|----------------|-------------------------|
| Sub-Index      | 2h                      |
| Description    | Alternate received PV 2 |
| Entry Category | Optional                |
| Access         | rw                      |
| PDO Mapping    | Possible                |
| Value Range    | Unsigned8               |
| Default Value  | No                      |

to

|                |                           |
|----------------|---------------------------|
| Sub-Index      | C7h                       |
| Description    | Alternate received PV 199 |
| Entry Category | Optional                  |
| Access         | rw                        |
| PDO Mapping    | Possible                  |
| Value Range    | Unsigned8                 |
| Default Value  | No                        |

**7.2.131 Object 6F52h: Received status**

General input status value which is used as input for analogue output (output process value), controller (effective current value) or alarm (input value for alarm 1..16) function block depending on type of module. This status byte will make sure that input process values are only used in these functional blocks when they are valid.

## OBJECT DESCRIPTION

|             |                        |
|-------------|------------------------|
| Index       | 6F52h                  |
| Name        | <b>Received status</b> |
| Object Code | ARRAY                  |
| Data Type   | Unsigned8              |
| Category    | Mandatory              |

## ENTRY DESCRIPTION

|                |                    |
|----------------|--------------------|
| Sub-Index      | 0h                 |
| Description    | number of channels |
| Entry Category | Mandatory          |
| Access         | ro                 |
| PDO Mapping    | No                 |
| Value Range    | 1 - C7h            |
| Default Value  | No                 |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 1h                       |
| Description    | <b>Received status 1</b> |
| Entry Category | Mandatory                |
| Access         | rw                       |
| PDO Mapping    | Possible                 |
| Value Range    | Unsigned8                |
| Default Value  | No                       |

|                |                          |
|----------------|--------------------------|
| Sub-Index      | 2h                       |
| Description    | <b>Received status 2</b> |
| Entry Category | Optional                 |
| Access         | rw                       |
| PDO Mapping    | Possible                 |
| Value Range    | Unsigned8                |
| Default Value  | No                       |

to

|                |                            |
|----------------|----------------------------|
| Sub-Index      | C7h                        |
| Description    | <b>Received status 199</b> |
| Entry Category | Optional                   |
| Access         | rw                         |
| PDO Mapping    | Possible                   |
| Value Range    | Unsigned8                  |
| Default Value  | No                         |

**7.2.132 Object 6F53h: Alternate received status**

General input status value, which is used as alternate input for analogue output (output process value), controller (effective current value) or alarm (input value for alarm 1..16) function block depending on type of module. This status byte will make sure that alternate input process values are only used in these functional blocks when they are valid.

## OBJECT DESCRIPTION

|             |                                  |
|-------------|----------------------------------|
| Index       | 6F53h                            |
| Name        | <b>Alternate received status</b> |
| Object Code | ARRAY                            |
| Data Type   | Unsigned8                        |
| Category    | Optional                         |

## ENTRY DESCRIPTION

|                |                    |
|----------------|--------------------|
| Sub-Index      | 0h                 |
| Description    | number of channels |
| Entry Category | Mandatory          |
| Access         | ro                 |
| PDO Mapping    | No                 |
| Value Range    | 1 - C7h            |
| Default Value  | No                 |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 1h                                 |
| Description    | <b>Alternate received status 1</b> |
| Entry Category | Mandatory                          |
| Access         | rw                                 |
| PDO Mapping    | Possible                           |
| Value Range    | Unsigned8                          |
| Default Value  | No                                 |

|                |                                    |
|----------------|------------------------------------|
| Sub-Index      | 2h                                 |
| Description    | <b>Alternate received status 2</b> |
| Entry Category | Optional                           |
| Access         | rw                                 |
| PDO Mapping    | Possible                           |
| Value Range    | Unsigned8                          |
| Default Value  | No                                 |

to

|                |                                      |
|----------------|--------------------------------------|
| Sub-Index      | C7h                                  |
| Description    | <b>Alternate received status 199</b> |
| Entry Category | Optional                             |
| Access         | rw                                   |
| PDO Mapping    | Possible                             |
| Value Range    | Unsigned8                            |
| Default Value  | No                                   |

**7.2.133 Object 6F60h: Transmission Enable**

This object enables transmission of PDOs when a transmission event is detected. This object may be used to make sure that all receiving modules are ready within the operational state before PDOs are transmitted within the bus.

## OBJECT DESCRIPTION

|             |                     |
|-------------|---------------------|
| Index       | 6F60h               |
| Name        | Transmission Enable |
| Object Code | VAR                 |
| Data Type   | Boolean             |
| Category    | Optional            |

ENTRY DESCRIPTION

|               |         |
|---------------|---------|
| Access        | rw      |
| PDO Mapping   | No      |
| Value Range   | Boolean |
| Default Value | FALSE   |